



Chapter 5

Simple mixtures

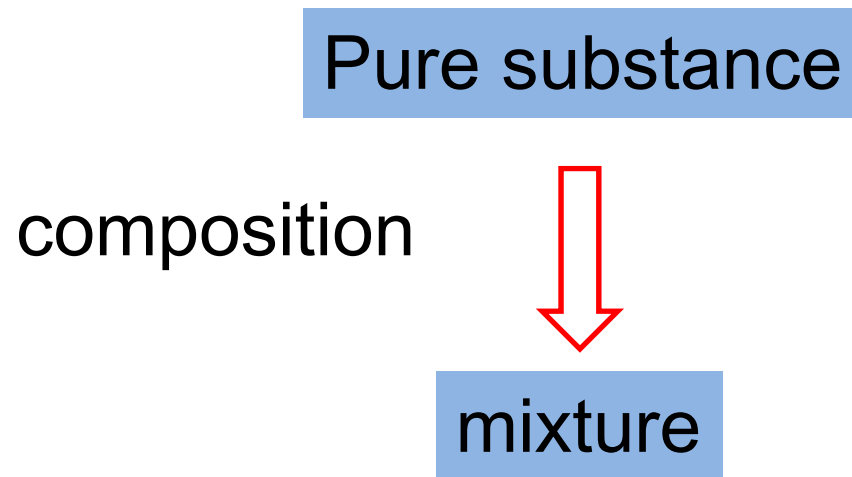


Outline

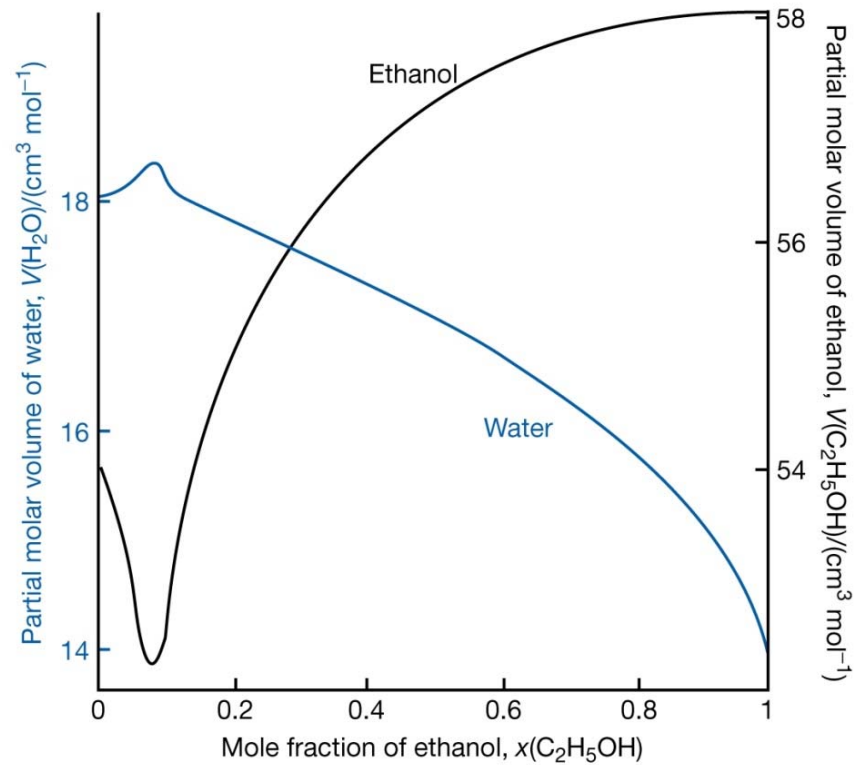
- The thermodynamic description of mixture
- The chemical potentials of solutions
 - Raoult's law
 - Henry's law
- The properties of solutions
- The real solution (activity)



5.1 Partial molar quantities



$$\text{state function} = f(T, p)$$
$$\text{state function} = f(T, p, \text{composition})$$



The reason for the different increase in volume is that the volume occupied by a given number of water molecules depends on the identity of the molecules that surround them.



Simple mixture

Mixture

- ❑ Multiple components
- ❑ Homogeneous phase
- ❑ Components change in a specific range

Simple

- ❑ No chemical reactions
- ❑ Non-electrolyte solutions
- ❑ A binary mixture $x_A + x_B = 1$



1. Composition ($x_A + x_B = 1$)

□ Mole fraction of B

$$x_B = \frac{n_B}{n_A + n_B}$$

□ mass fraction of B

$$w_B = \frac{m_B}{m_A + m_B}$$

□ Mole concentration of B

$$C_B = \frac{n_B}{V} \quad \text{mol}/\text{m}^3$$

□ Molality 质量摩尔浓度 of B

$$b_B = \frac{n_B}{m_A} \quad \text{mol}/\text{kg}$$

□ Mass concentration of B

$$\rho_B = \frac{m_B}{V} \quad \text{kg}/\text{m}^3$$



2. Partial molar quantities

$$X = f(T, p, n_A, n_B)$$

$$dX = \left(\frac{\partial X}{\partial T} \right)_{p, n_A, n_B} dT + \left(\frac{\partial X}{\partial p} \right)_{T, n_A, n_B} dp + \left(\frac{\partial X}{\partial n_A} \right)_{T, p, n_B} dn_A + \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A} dn_B$$

Constant T and p

$$dX = \left(\frac{\partial X}{\partial n_A} \right)_{T, p, n_B} dn_A + \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A} dn_B$$



$$\left(\frac{\partial X}{\partial n_A} \right)_{T, p, n_B} = X_A \quad \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A} = X_B$$

$$dX = X_A dn_A + X_B dn_B$$



$$X_B = \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A}$$

Molar quantity

- For pure substance, the quantity at one molar.

Partial molar quantity

- The contribution that a component of a mixture makes to the total quantity.



Definition

$$X_B = \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A} \quad \text{Conditions} \left\{ \begin{array}{l} 1. \text{ homogeneous phase} \\ 2. T, p, n_A \text{ constant} \\ 3. +/- \end{array} \right.$$

X : state function
extensive properties $f(T, p, n_A, n_B)$

X_B : State function
intensive properties $f(T, p, x_A)$

$$dX = X_A dn_A + X_B dn_B$$



$$V_B = \left(\frac{\partial V}{\partial n_B} \right)_{T, p, n_A}$$

$$U_B = \left(\frac{\partial U}{\partial n_B} \right)_{T, p, n_A}$$

$$H_B = \left(\frac{\partial H}{\partial n_B} \right)_{T, p, n_A}$$

$$S_B = \left(\frac{\partial S}{\partial n_B} \right)_{T, p, n_A}$$

$$A_B = \left(\frac{\partial A}{\partial n_B} \right)_{T, p, n_A}$$

$$G_B = \left(\frac{\partial G}{\partial n_B} \right)_{T, p, n_A}$$



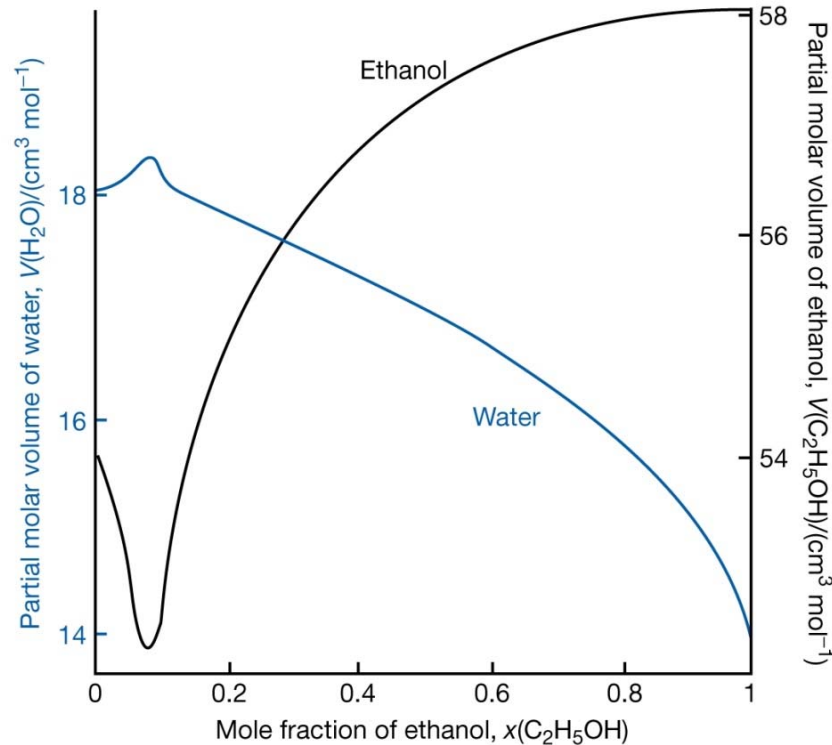
a. Partial molar volume

$$V_B = \left(\frac{\partial V}{\partial n_B} \right)_{T, p, n_A}$$

- General condition $V_B \neq V_{m,B}$
- Pure substance $V_B = V_{m,B}$
- $x_B \uparrow$ $V_B \rightarrow V_{m,B}$
- $x_B \rightarrow 0$ $V_A \rightarrow V_{m,A}$



Water & Ethanol



1 mol water is added to a huge volume of ethanol:

$$V_{\text{H}_2\text{O}} = \sim 14 \text{ cm}^3 \text{mol}^{-1}$$

$$V_{\text{H}_2\text{O}, m} = 18 \text{ cm}^3 \text{mol}^{-1}$$

$(V_{\text{ethanol}}, V_{\text{H}_2\text{O}})$ change with compositions.

V_B might be negative

1 mol MgSO_4 is added to a huge volume of very diluted MgSO_4 water solution: $V_B = -1.4 \text{ cm}^3 \text{mol}^{-1}$

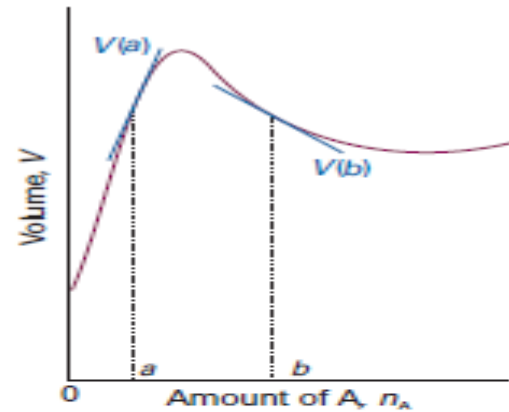


Fig. 5.2 The partial molar volume of a substance is the slope of the variation of the total volume of the sample plotted against the composition. In general, partial molar quantities vary with the composition, as shown by the different slopes at the compositions *a* and *b*. Note that the partial molar volume at *b* is negative: the overall volume of the sample decreases as *A* is added.

$$X_B = \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A}$$

The partial molar volumes of the components of a mixture vary with composition because the environment of each type of molecule changes as the composition changes from pure A to pure B. It is this changing molecular environment, and the consequential modification of the forces acting between molecules, that results in the variation of the thermodynamic properties of a mixture as its composition is changed.



Additive formula

- Changes of state (composition changes)

$$dX = X_A dn_A + X_B dn_B$$

- One state at certain T, p (with certain n_A and n_B)

$$X = \int_0^{n_A} X_A dn_A + \int_0^{n_B} X_B dn_B$$

At one state

$$(n_A, n_B) \Rightarrow (x_A, x_B) \Rightarrow (X_A, X_B)$$

$$X = X_A n_A + X_B n_B$$



Additive formula (at a given composition)

$$X = \sum_{i=1}^k n_i X_i$$

$$X = n_A X_A + n_B X_B$$

$$V = n_A V_A + n_B V_B$$

$$U = n_A U_A + n_B U_B$$

$$H = n_A H_A + n_B H_B$$

$$S = n_A S_A + n_B S_B$$

$$A = n_A A_A + n_B A_B$$

$$G = n_A G_A + n_B G_B$$



5.1(a) The partial molar volumes of acetone (propanone) and chloroform (trichloromethane) in a mixture in which the mole fraction of CHCl_3 is 0.4693 are $74.166 \text{ cm}^3 \text{ mol}^{-1}$ and $80.235 \text{ cm}^3 \text{ mol}^{-1}$, respectively. What is the volume of a solution of mass 1.000 kg?



5.1(a) The partial molar volumes of acetone (propanone) and chloroform (trichloromethane) in a mixture in which the mole fraction of CHCl_3 is 0.4693 are $74.166 \text{ cm}^3 \text{ mol}^{-1}$ and $80.235 \text{ cm}^3 \text{ mol}^{-1}$, respectively. What is the volume of a solution of mass 1.000 kg?

$$M(\text{acetone}) = 58.08$$
$$M(\text{chloroform}) = 119.38$$

$$V_a = 74.166 \text{ cm}^3 \text{ mol}^{-1}$$

$$V_c = 80.235 \text{ cm}^3 \text{ mol}^{-1}$$

$$x_c = 0.4693 \quad x_a = 0.5307$$

$$V = V_a \times n_a + V_c \times n_c \quad \text{混合公式}$$

$$m = n \times x_a \times M_a + n \times x_c \times M_c$$

$$m = n (M_a \cdot x_a + M_c \cdot x_c)$$

$$1 \text{ kg} = n (58.08 \times 0.5307 + 119.38 \times 0.4693)$$

$$1 = n (30.823 + 56.025)$$

$$1000 \text{ g} = n \times 86.848 \text{ g/mol}$$

$$n = 11.514 \text{ mol}$$

$$n_a = 11.514 \times 0.5307 = 6.11 \text{ mol}$$

$$n_c = 11.514 \times 0.4693 = 5.40 \text{ mol}$$

$$V = V_a \times n_a + V_c \times n_c$$
$$= 74.166 \times 6.11 + 80.235 \times 5.40$$
$$= 453.15 + 433.269$$
$$= 886.42 \text{ cm}^3$$



b. Partial molar Gibbs energies

For a pure substance:

$$\mu = G_m = \left(\frac{\partial G}{\partial n} \right)_{p, T}$$

For a substance in a mixture:

$$\mu_B = \left(\frac{\partial G}{\partial n_B} \right)_{T, p, n_A}$$



In an open system of constant composition, the Gibbs energy depends on the n_i , p , and T . Thus, G may change when p , T , and the n_i .

$$dG = \left(\frac{\partial G}{\partial T}\right)_{p, n_A, n_B} dT + \left(\frac{\partial G}{\partial p}\right)_{T, n_A, n_B} dp + \left(\frac{\partial G}{\partial n_A}\right)_{T, p, n_B} dn_A + \left(\frac{\partial G}{\partial n_B}\right)_{T, p, n_A} dn_B$$

And $dG = Vdp - SdT$ becomes:

$$dG = Vdp - SdT + \mu_A dn_A + \mu_B dn_B$$

At constant p and T

$$dG = \mu_A dn_A + \mu_B dn_B$$



Since $G=U+pV-TS$,

$$dU = -pdV - Vdp + SdT + TdS + dG$$

$$= -pdV - \cancel{Vdp} + \cancel{SdT} + TdS + (\cancel{Vdp} - \cancel{SdT} + \mu_A dn_A + \mu_B dn_B + \dots)$$

$$dU = -pdV + TdS + \mu_A dn_A + \mu_B dn_B$$

At constant V and S

$$dU = \mu_A dn_A + \mu_B dn_B$$

$$\mu_B = \left(\frac{\partial U}{\partial n_B} \right)_{S, V, n_A}$$



$$dG = Vdp - SdT + \mu_A dn_A + \mu_B dn_B$$

$$dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B$$

$$dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B$$

$$dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B$$

Partial molar quantity

$$\mu_B = \left(\frac{\partial G}{\partial n_B} \right)_{T, p, n_A}$$

$$\mu_B = \left(\frac{\partial U}{\partial n_B} \right)_{S, V, n_A}$$

$$\mu_B = \left(\frac{\partial H}{\partial n_B} \right)_{S, p, n_A}$$

$$\mu_B = \left(\frac{\partial A}{\partial n_B} \right)_{T, V, n_A}$$

The results show how all the extensive thermodynamic properties U, H, A and G depend on the composition.



Equilibrium criterion of chemical potential

$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i$$

<0 Spontaneous
=0 Equilibrium

The driving force for mass transfer is typically a difference in chemical potential.



3. The Gibbs-Duhem equation

The total Gibbs energy of a mixture depends on μ and n , for a binary mixture:

$$G = n_A \mu_A + n_B \mu_B$$

$$dG = \mu_A dn_A + \mu_B dn_B + n_A d\mu_A + n_B d\mu_B$$

At constant p and T

$$dG = \mu_A dn_A + \mu_B dn_B$$

So $n_A d\mu_A + n_B d\mu_B = 0$ Gibbs-Duhem equation



The Gibbs-Duhem equation

$$\sum_J n_J d\mu_J = 0$$

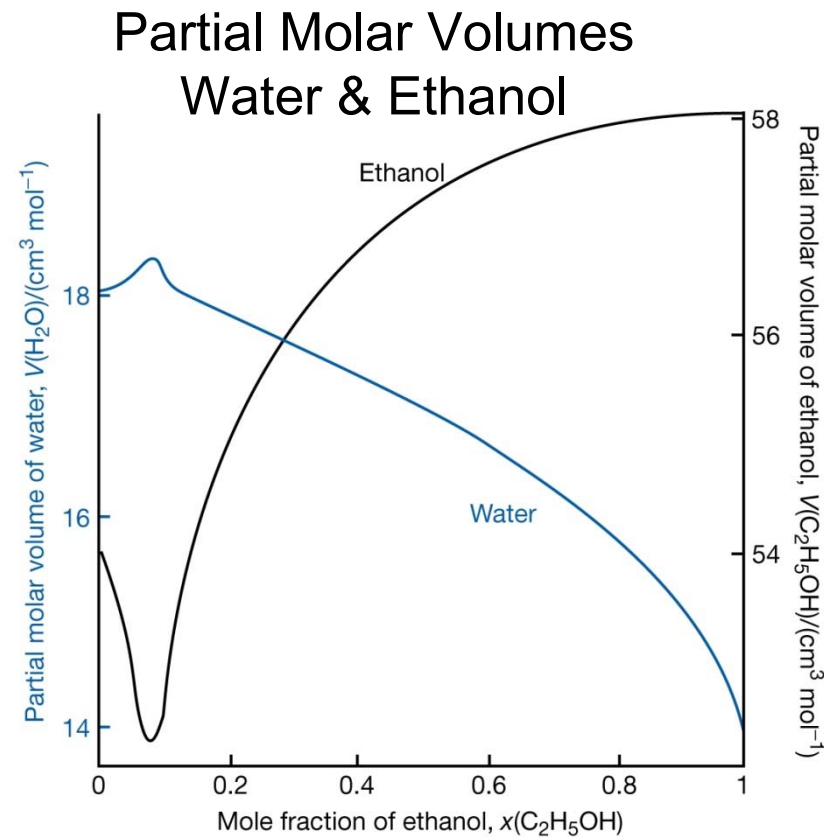
The significance of the Gibbs-Duhem equation is that the chemical potential of one component of a mixture cannot change independently of the chemical potentials of the other components.

In general
$$\sum_J n_J dX_J = 0$$



In a binary mixture, if one partial molar quantity increases, the other must decrease:

$$dV_B = -\frac{n_A}{n_B} dV_A$$



This is true for all partial molar quantities.



Example

The experimental value of the partial molar volume of $\text{K}_2\text{SO}_4(\text{aq})$ at 298 K is given by the expression

$$V_{\text{K}_2\text{SO}_4}/(\text{cm}^3\text{mol}^{-1}) = 32.280 + 18.216b^{1/2}$$

where b is the numerical value of the molality of K_2SO_4 . Use the Gibbs-Duhem equation to derive an equation for the molar volume of water in the solution. The molar volume of pure water at 298 K is $18.079 \text{ cm}^3\text{mol}^{-1}$



Method: Let A denote K_2SO_4 and B denote H_2O , the solvent. The Gibbs-Duhem equation for the partial molar volumes of two components is

$$\sum_J n_J dX_J = 0 \quad n_A dX_A + n_B dX_B = 0$$

this relation implies that

$$dV_B = -\frac{n_A}{n_B} dV_A \quad \longrightarrow \quad V_B = V_B^* - \int \frac{n_A}{n_B} dV_A$$

Change the variable V_A to the molality b , and integrate the right-hand side between $b = 0$ (pure B) and the molality of interest.

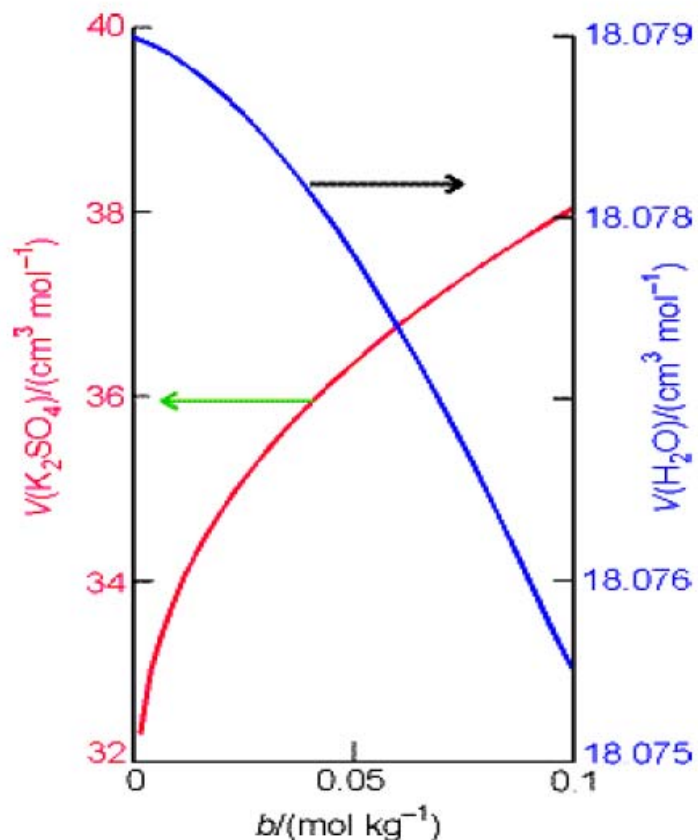


Answer: It follows from the information in the question, that, with $A = K_2SO_4$, and $V_A = 32.280 + 18.216b^{1/2}$

$$\frac{dV_A}{db} = 9.108 b^{-1/2}$$

$$V_B = V_B^* - \int \frac{n_A}{n_B} dV_A \quad \longrightarrow \quad V_B = V_B^* - 9.108 \int \frac{n_A}{n_B} b^{-1/2} db$$

$$b = \frac{n_A}{n_B M_B}, \quad \text{where } b \text{ is the molar mass of water}$$



The partial molar volumes of the components of an aqueous solution of potassium sulfate.

$$V_B = V_B^* - 9.108 M_B \int b^{1/2} db$$

$$= V_B^* - \frac{2}{3} (9.108 M_B b^{3/2})$$

$$V_B = V_B^* - \frac{2}{3} (9.108 M_B b^{3/2})$$

By substituting the data

$$V_B / (\text{cm}^3 \text{mol}^{-1}) = 18.079 - 0.1094 b^{3/2}$$



4. The Gibbs energy of gas mixtures

a. The Gibbs energy of pure gases

$$dG = Vdp - SdT$$

Constant T

$$\left(\frac{\partial G_m}{\partial p}\right)_T = V_m = \frac{RT}{p} \quad G_{m,f} - G_{m,i} = RT \int_{p_i}^{p_f} \frac{dp}{p} \\ = RT \int_{p_i}^{p_f} d \ln p$$

$p_i = 1 \text{ bar}$; pure substance

$$G_{m,i} = \mu^\ominus; G_{m,f} = \mu$$

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus} \quad \mu^\ominus = f(T)$$



b. Gas mixtures

Dalton's law---the perfect gases

The pressure exerted by a mixture of gases is the sum of the partial pressures of the gases.

$B(P_B), C(P_C)$
.....(k)
Mixture (P)

$$P_B + P_C + \dots P_i = P$$

$$P_B = \frac{n_B RT}{V}$$

$$P_C = \frac{n_C RT}{V}$$

⋮

$$P_i = \frac{n_i RT}{V}$$

$$\frac{P_i}{P} = \frac{n_i}{n} = x_i$$

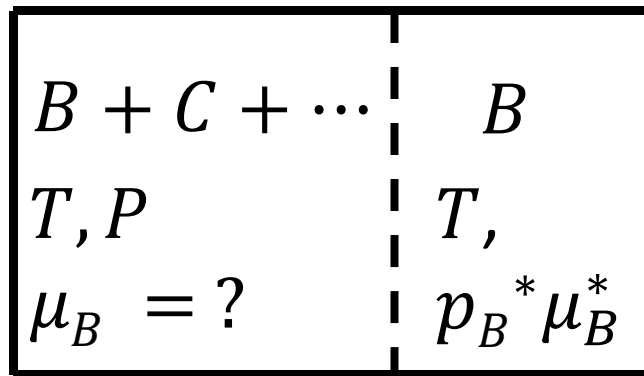
$$\sum_{i=1}^k P_i = \frac{RT}{V} \sum_{i=1}^k n_i$$

$$P = \frac{nRT}{V}$$

$$P_i = x_i P$$



An ideal gas mixture separated from pure gas B by a membrane permeable to B only



Semipermeable membrane

$$p_B = p_B^* \quad p_B = x_B P$$

$$\mu_B(T, P, x_1, x_2 \dots) = \mu_B^*(T, p_B^*)$$

$$\mu_B^* = \mu_B^\ominus(T) + RT \ln \frac{p_B^*}{p_B^\ominus}$$

$$\mu_B = \mu_B^\ominus(T) + RT \ln \frac{p_B}{p_B^\ominus}$$

$$\mu_B = \mu_B^\ominus(T) + RT \ln \frac{p_B}{p^\ominus}$$

Partial pressure

1 bar



$$\left(\frac{\partial \mu_i}{\partial P}\right)_{T,n} = \left(\frac{\partial V}{\partial n_i}\right)_{T,P,n} \neq n_i$$

$$\left(\frac{\partial V}{\partial n_i}\right)_{T,P,n \neq n_i} = \left[\frac{\partial(nRT/P)}{\partial n_i}\right]_{T,P,n \neq n_i} = \frac{RT}{P} = x_i \frac{RT}{P_i}$$

$$\left(\frac{\partial \mu_i}{\partial P}\right)_{T,n} = \left(\frac{\partial \mu_i}{\partial P_i}\right)_{T,n} \left(\frac{\partial P_i}{\partial P}\right)_{T,n} = \left(\frac{\partial \mu_i}{\partial P_i}\right)_{T,n} \left[\frac{\partial(x_i P)}{\partial P}\right]_{T,n} = x_i \left(\frac{\partial \mu_i}{\partial P_i}\right)_{T,n}$$

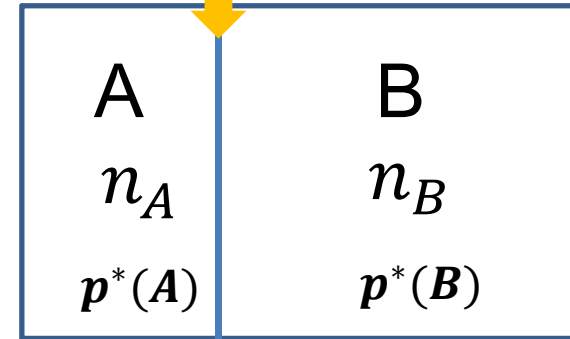
$$x_i \left(\frac{\partial \mu_i}{\partial P_i}\right)_{T,n} = x_i \frac{RT}{P_i}$$

$$\int_{\mu_i^\ominus}^{\mu_i} d\mu_i = \int_{P_i^\ominus}^{P_i} \left(\frac{\partial \mu_i}{\partial P_i}\right)_{T,n} dP_i = \int_{P_i^\ominus}^{P_i} \frac{RT}{P_i} dP_i$$

$$\mu_i = \mu_i^\ominus(T) + RT \ln \frac{p_i}{p_i^\ominus}$$



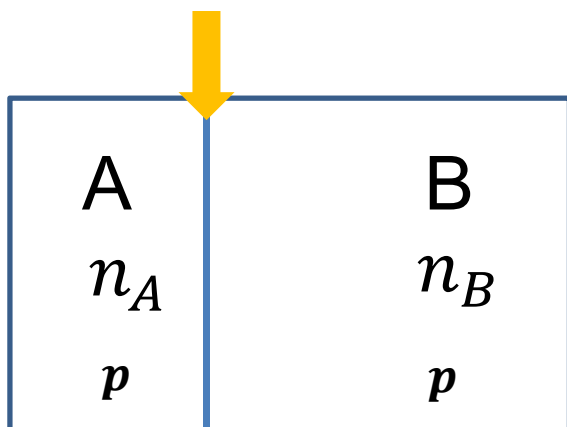
c. The Gibbs energy of mixing



$$\begin{aligned} G_i &= n_A \mu_A^* + n_B \mu_B^* \\ &= n_A (\mu_A^\ominus + RT \ln \frac{p^*(A)}{p_A^\ominus}) + n_B (\mu_B^\ominus + RT \ln \frac{p^*(B)}{p_B^\ominus}) \end{aligned}$$

$$\begin{aligned} G_f &= n_A \mu_A + n_B \mu_B \\ &= n_A (\mu_A^\ominus + RT \ln \frac{p_A}{p_A^\ominus}) + n_B (\mu_B^\ominus + RT \ln \frac{p_B}{p_B^\ominus}) \end{aligned}$$

$$\begin{aligned} \Delta G_{mix} &= G_f - G_i \\ &= n_A RT \ln \frac{p_A}{p^*(A)} + n_B RT \ln \frac{p_B}{p^*(B)} \end{aligned}$$



Dalton's law

$$(n_A + n_B)RT = p_f(V_A + V_B)$$

Before mixing

$$n_A RT = p V_A$$

$$n_B RT = p V_B$$

$$\therefore p_f = p$$

$$p^*(A) = p^*(B) = p$$

$$\Delta G_{mix} = n_A RT \ln \frac{p_A}{p^*(A)} + n_B RT \ln \frac{p_B}{p^*(B)}$$

$$\Delta G_{mix} = n_A RT \ln \frac{p_A}{p} + n_B RT \ln \frac{p_B}{p}$$

$$\Delta G_{mix} = n_A RT \ln x_A + n_B RT \ln x_B$$

$$\Delta G_{mix} = nRT(x_A \ln x_A + x_B \ln x_B)$$

$$\ln x_A < 0 \quad \ln x_B < 0$$

$$\Delta G_{mix} < 0$$

The perfect gas mixing is spontaneous in all proportions.



$$dG = Vdp - SdT$$

Constant p

$$\left(\frac{\partial G}{\partial T}\right)_{p,n_A,n_B} = -S$$

$$\Delta_{mix}S = -\left(\frac{\partial \Delta_{mix}G}{\partial T}\right)_{p,n_A,n_B} = -nR(x_A \ln x_A + x_B \ln x_B)$$

$$\Delta_{mix}S = -nR(x_A \ln x_A + x_B \ln x_B) > 0$$

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta_{mix}H = 0$$

The driving force for mixing comes from the increase in entropy of the system, because the entropy of the surroundings is unchanged.



$$\Delta_{mix}S > 0$$

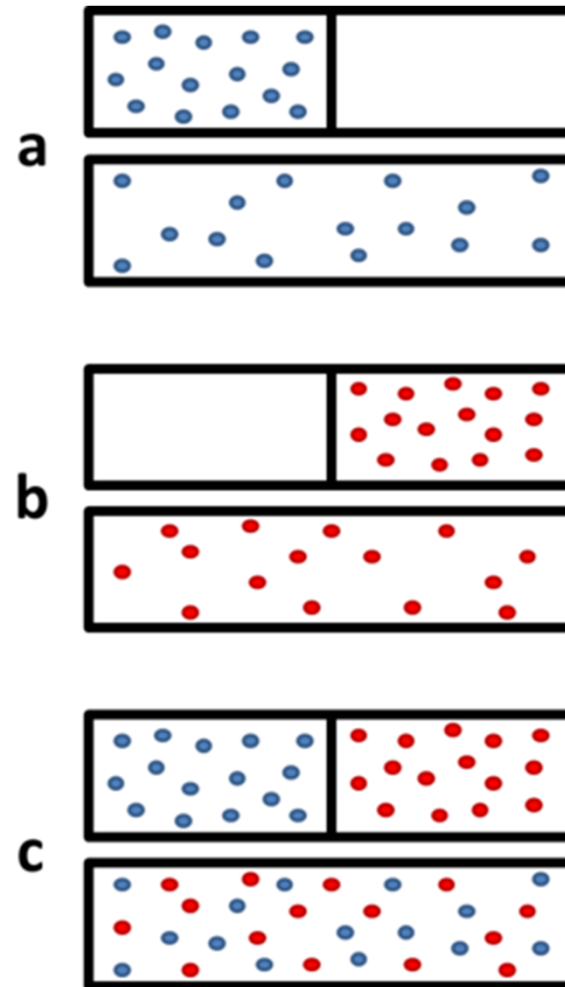


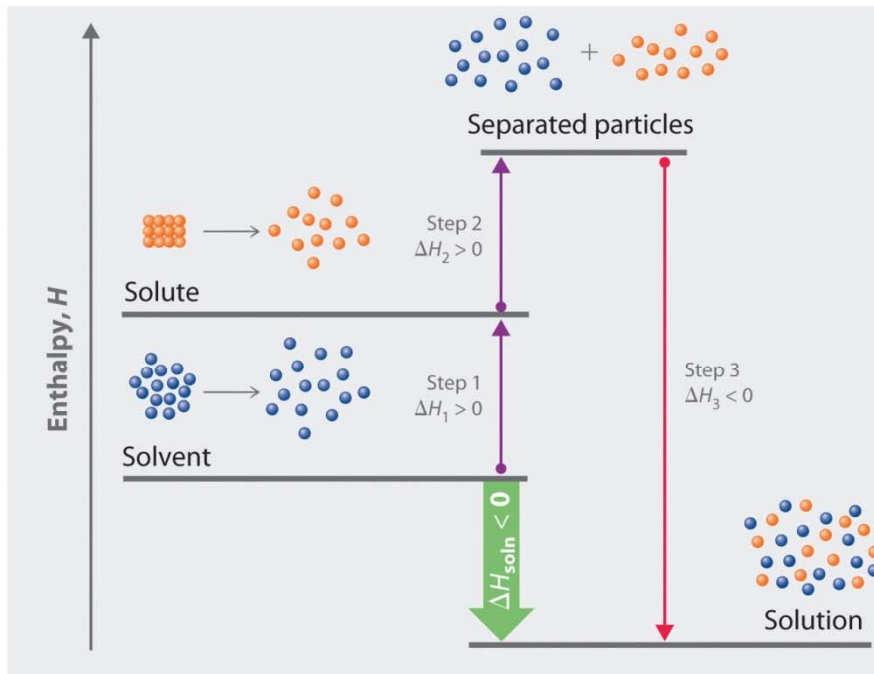
Figure 1. The mixing of two gases can be seen as two expansions.

- a) Expansion of gas 1 alone when the barrier is removed. The molecules have twice as many microstates in the open box.
- b) Expansion of gas 2 alone.
- c) The simultaneous expansion of gases 1 and 2 is equivalent to mixing.

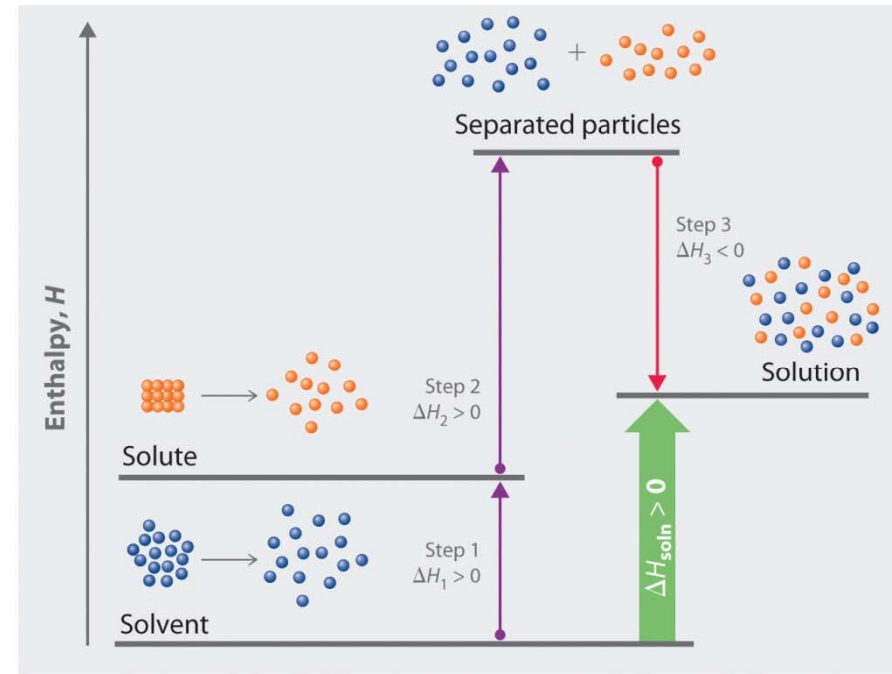
When one gas disperses into the other and the system becomes more disordered.



$$\Delta_{mix}H = 0$$

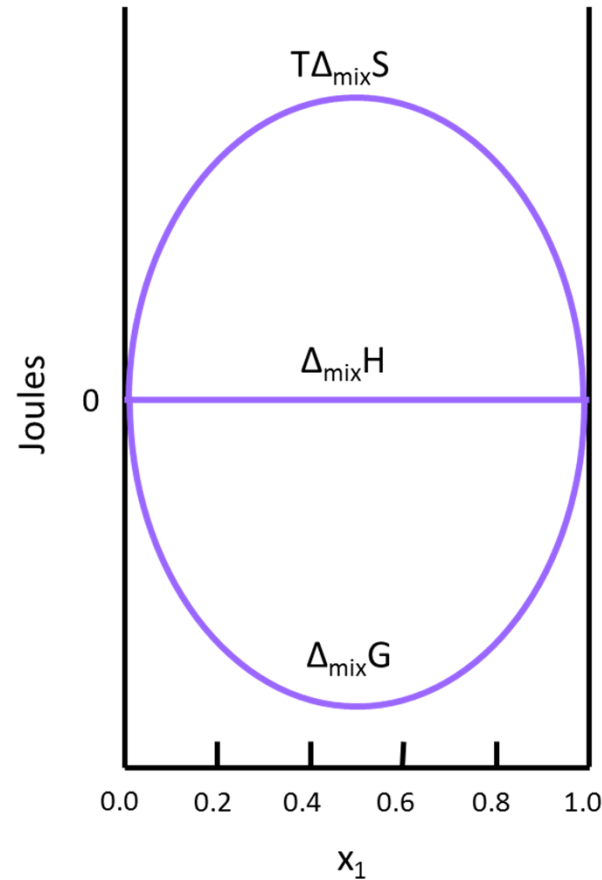


(a) Exothermic solution formation



(b) Endothermic solution formation

The enthalpy of mixing is zero because there are no interactions between the molecules forming the gaseous mixture.



$$\Delta_{\text{mix}}S > 0$$

$$\Delta_{\text{mix}}H = 0$$

$$\Delta_{\text{mix}}G < 0$$

Figure 2. A graph of $T\Delta_{\text{mix}}S$, $\Delta_{\text{mix}}H$, and $\Delta_{\text{mix}}G$ as functions of x_1 for the mixing of two ideal gases.

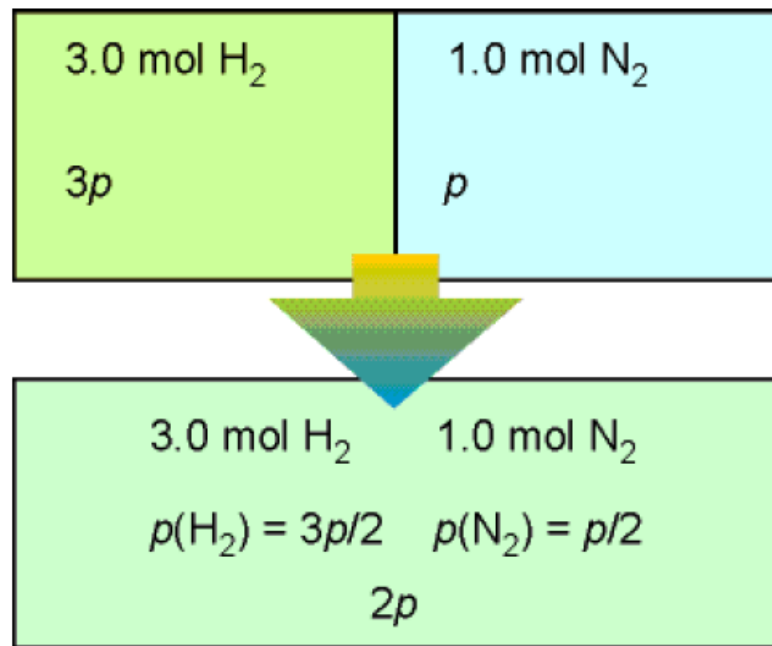


Calculating a Gibbs energy of mixing

A container is divided into two equal compartments. One contains 3.0 mol H_2 at 25 °C; the other contains 1.0 mol N_2 at 25 °C. Calculate the Gibbs energy of mixing when the partition is removed. Assume perfect behavior.



Method: We proceed by calculating the initial Gibbs energy from the chemical potentials. To do so, we need the pressure of each gas. Write the pressure of nitrogen as p ; then the pressure of hydrogen as a multiple of p



can be found from the gas laws. Next, calculate the Gibbs energy for the system when the partition is removed. The volume of each gas doubles, so its partial pressure falls by a factor of 2.



Answer: Given that the pressure of N_2 is p , the pressure of H_2 is $3p$; therefore, the initial Gibbs energy is

$$\begin{aligned} G_1 &= n_A \left\{ \mu_A^\ominus + RT \ln p \right\} + n_B \left\{ \mu_B^\ominus + RT \ln p \right\} \\ &= (3.00 \text{ mol}) \left\{ \mu_A^\ominus (\text{H}_2) + RT \ln 3p \right\} + (1.00 \text{ mol}) \left\{ \mu_B^\ominus (\text{N}_2) + RT \ln p \right\} \end{aligned}$$

The partial pressure of N_2 falls to $\frac{1}{2}p$ and that of H_2 falls to $\frac{3}{2}p$. Therefore, the Gibbs energy changes to

$$G_2 = (3.00 \text{ mol}) \left\{ \mu_A^\ominus (\text{H}_2) + RT \ln \frac{3}{2}p \right\} + (1.00 \text{ mol}) \left\{ \mu_B^\ominus (\text{N}_2) + RT \ln \frac{1}{2}p \right\}$$



The Gibbs energy of mixing is the difference of these two quantities:

$$\begin{aligned}\Delta_{\text{mix}}G &= (3.00\text{mol})RT \ln \frac{\frac{3}{2}p}{3p} + (1.00\text{mol})RT \ln \frac{\frac{1}{2}p}{p} \\ &= -(4.00\text{mol})RT \ln 2 = -6.9\text{kJ}\end{aligned}$$

In this example, the value of $\Delta_{\text{mix}}G$ is the sum of two contributions: the mixing itself, and the changes in pressure of the two gases to their final pressure, $2p$. When two gases mix at the same pressure, the change of Gibbs energy is only from mixing, that is -5.6 kJ .



5. The thermodynamic of liquids

Raoult's law and ideal solution

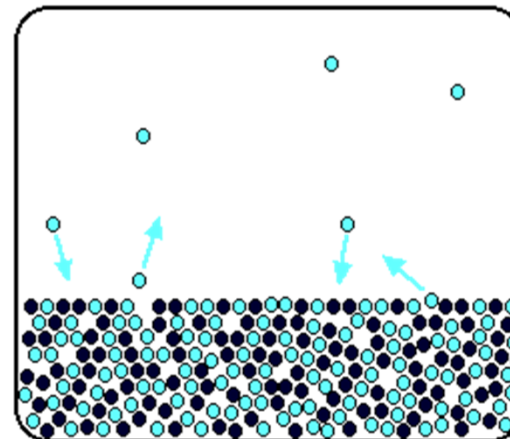
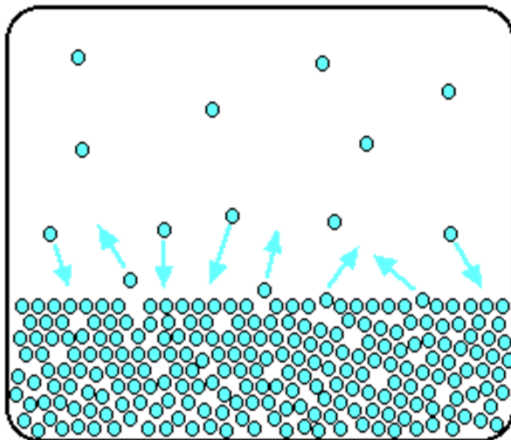
(1) Raoult's law

$$p_A = x_A p^*$$

p_A : the vapor pressure of component A of the solution

p^* : the vapour pressure as a pure liquid,

x_A : the mole fraction of A in the liquid mixture.





The Raoult's law reflects the fact that the presence of a second component reduces the rate at which A molecules leave the surface of the liquid but does not inhibit the rate at which they return.

The rate of evaporation is related to the number of molecules at the liquid surface.

$$v_{\text{evaporation}} = kx_A$$

The rate of condensation is related to the number of molecules in the gas phase.

$$v_{\text{condensation}} = k'p_A$$

In equilibrium,

$$v_{\text{evaporation}} = v_{\text{condensation}}$$



$$kx_A = k'p_A$$

$$p_A = \frac{k}{k'}x_A$$

$$\text{If } x_A=1; \quad p^* = \frac{k}{k'}$$

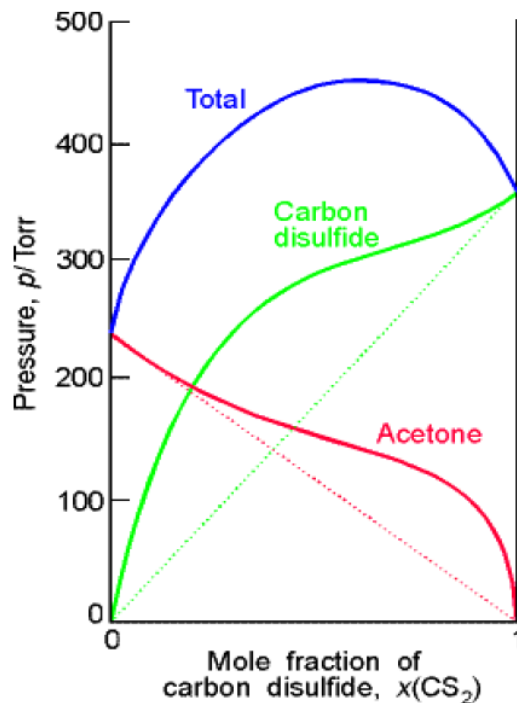
$$\therefore \quad \mathbf{p_A = x_A p^*}$$

Raoult's law tells that the ratio of the partial vapour pressure of each component to its vapour pressure as a pure liquid $\mathbf{p_A/p_A^*}$, is approximately equal to the mole fraction of A in the liquid mixture.



The limitations on Raoult's law

In practice, there's no such thing! However, very dilute solutions obey Raoult's Law to a reasonable approximation.



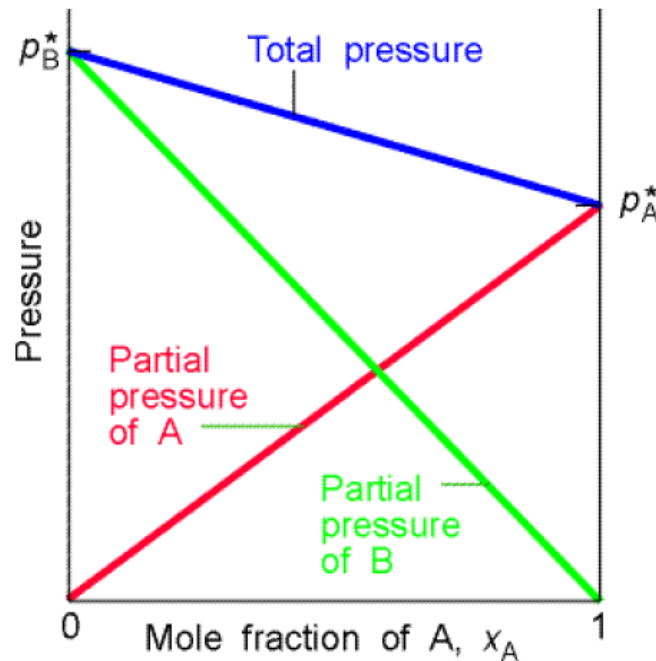
Some mixtures obey Raoult's law very well, especially when the components are **structurally similar**.

But some mixtures do not obey Raoult's law; and strong deviations from ideality are shown by dissimilar liquids, e.g. carbon disulfide and acetone.



(2) Ideal solution (a model)

Mixtures that obey the Raoult's law through out the composition range from pure A to pure B are called ideal solutions.





6. Ideal solutions and their chemical potential

$$\mu_B(l) = \mu_B(g)$$

$$\mu_B(g) = \mu^\ominus(g) + RT \ln \frac{p_B}{p_B^\ominus} = \mu_B(l)$$

$$p_B = x_B p^* \quad \mu_B(l) = \mu^\ominus(g) + RT \ln \frac{p_B^*}{p_B^\ominus} + RT \ln x_B$$

$$\text{If } x_B = 1 \quad \mu_B^*(l) = \mu^\ominus(g) + RT \ln \frac{p_B^*}{p_B^\ominus}$$

$$\mu_B(l) = \mu_B^*(l) + RT \ln x_B$$



The Gibbs energy of mixing of two liquids to form an ideal solution is calculated in exactly the same way as for two gases. The total Gibbs energy before liquids are mixed is:

$$G_i = n_A \mu_A^* + n_B \mu_B^*$$

when they are mixed, the individual chemical potential is:

$$\mu_i = \mu_i^* + RT \ln x_i$$

and the total Gibbs energy is:

$$G_f = n_A \{ \mu_A^* + RT \ln x_A \} + n_B \{ \mu_B^* + RT \ln x_B \}$$



The Gibbs energy of mixing, ΔG_{mix} , is

$$\Delta_{mix} G = G_f - G_i$$

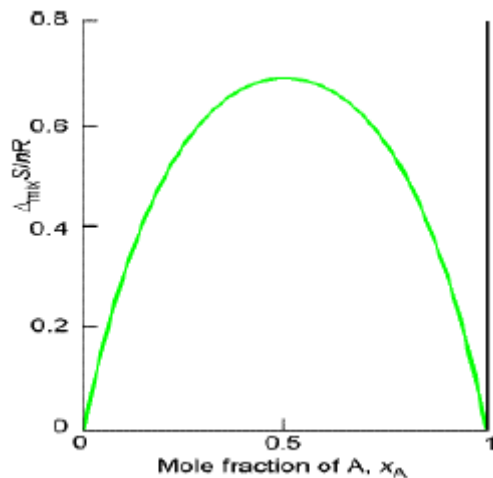
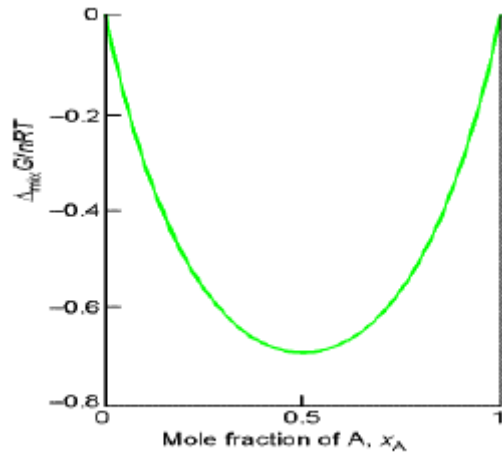
$$\Delta_{mix} G = nRT \{x_A \ln x_A + x_B \ln x_B\}$$

$$(\partial G / \partial T)_{p,n} = -S$$

It follows that the ideal entropy and enthalpy of mixing of two liquids is:

$$\Delta_{mix} S = -nR(x_A \ln x_A + x_B \ln x_B)$$

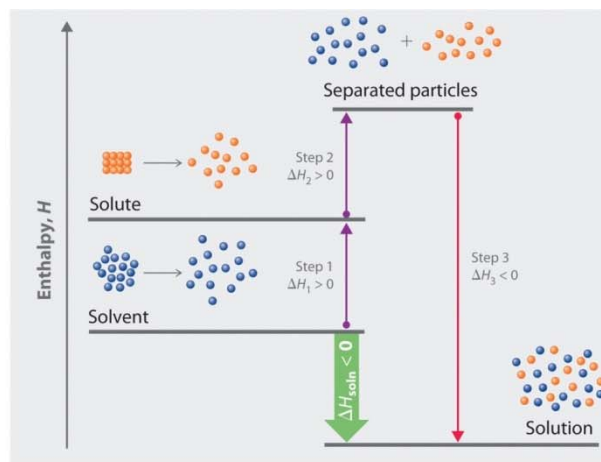
$$\Delta_{mix} H = 0$$



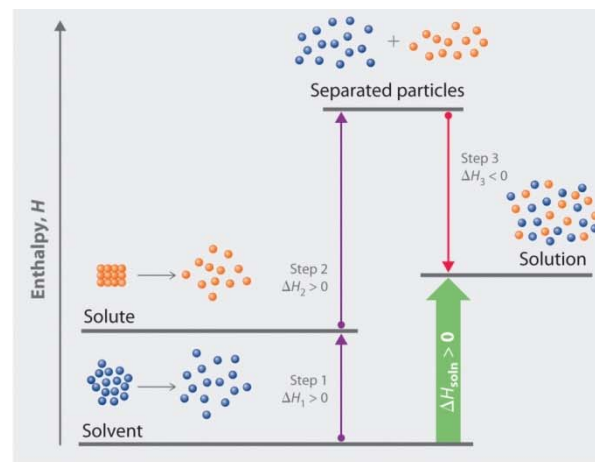
The variation of the Gibbs energy of mixing with composition is the same as that already depicted for gases (above); the same is true of the entropy of mixing (below). and all the conclusions drawn there are valid here: the driving force for mixing is the increasing entropy of the system as the molecules mingle, and the enthalpy of mixing is zero.



It should be noted that solution ideality means something different from gas perfection. In a perfect gas there are no interactions between molecules. In ideal solutions there are interactions, but the average A-B interactions in the mixture are the same as the average A-A and B-B interactions in the pure liquids.



(a) Exothermic solution formation



(b) Endothermic solution formation

$$\Delta_{mix}H = 0$$

$$\Delta_{mix}V = 0$$



7. Henry's law and ideal-dilute solutions

a. Henry's law

$$x_B \propto p_B$$

$$p_B = x_B K_B$$

x_B : the mole fraction of the solute

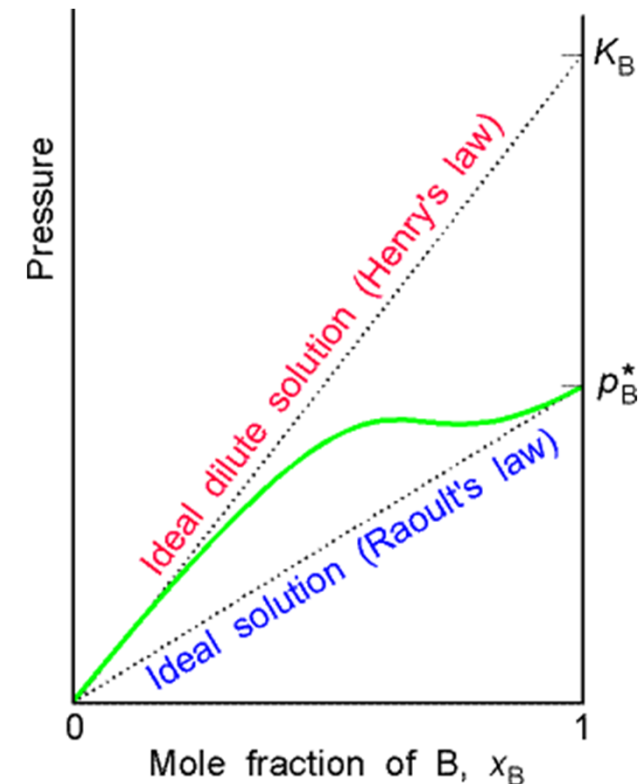
p_B : the vapour pressure of the solute

K_B : an empirical constant

$$K_B \neq p_B^*$$

How to determine K_B ?

the tangent to the experimental curve of plot of the vapour pressure of B against its mole fraction at $x_B = 0$.



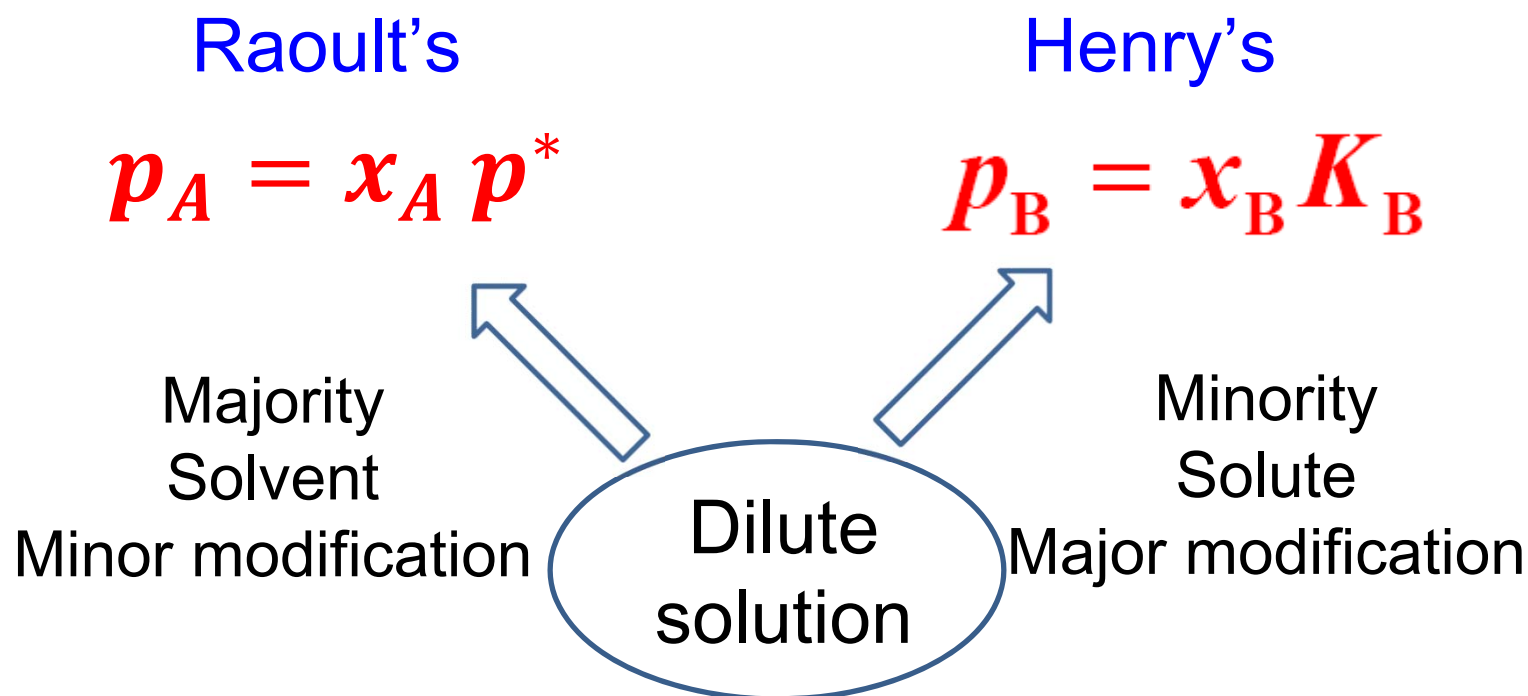


$$p_B = x_B K_B$$

- Henry found that for real solutions at low concentrations
- $K = f(T, P, \text{solute}, \text{solvent})$
- The law cannot be applied to electrolytes.
- An empirical constant with the dimensions of pressure.
- Be used to calculate gas solubility.



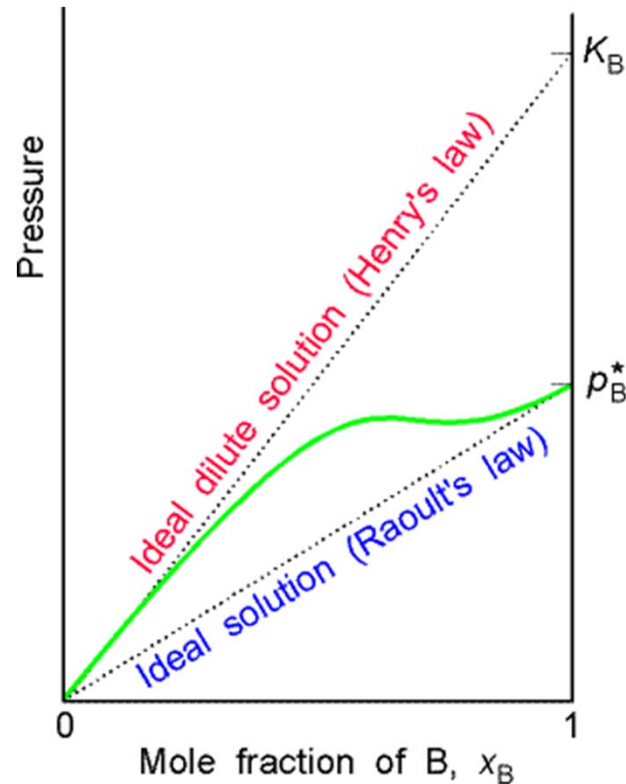
The difference and relations between Raoult's law and Henry's law



In general, however, both Raoults and Henrys laws are limit laws, applying at opposite ends of the composition range.



b. ideal-dilute solutions



- Mixtures for which the solute obeys Henry's law and the solvent obeys Raoult's law are called ideal-dilute solutions.
- Henry's law applies to the **solute** in ideal dilute solutions.
- Raoult's law applies to **solvent** in ideal dilute solutions.
- Real systems can (and do) deviate from both



Investigating the validity of Raoult's and Henry's laws

The vapour pressures of each component in a mixture of acetone(A) and chloroform(C) were measured at 35 °C with the following results:

x_C	0	0.20	0.40	0.60	0.80	1
p_C/Torr	0	35	82	142	219	293
p_A/Torr	347	270	185	102	37	0

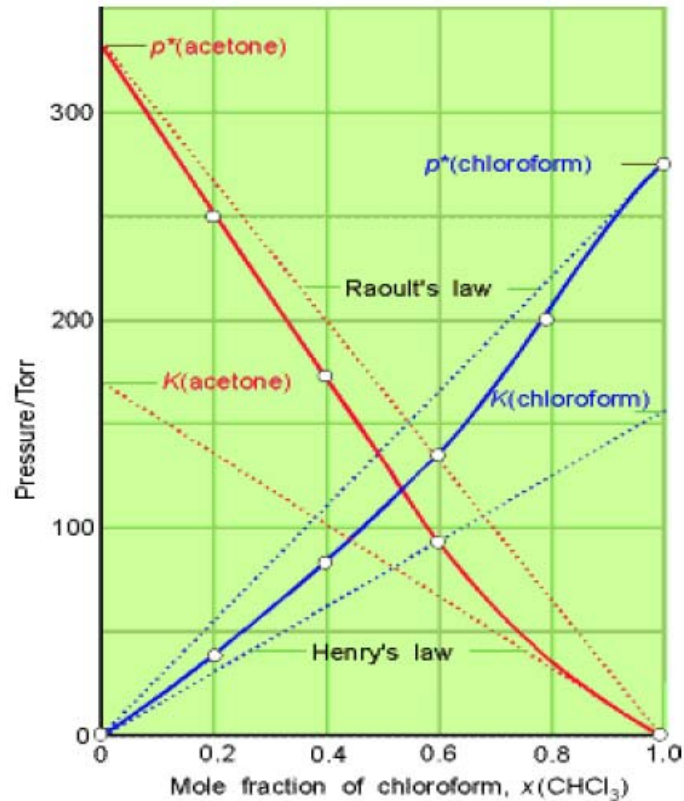
Confirm that the mixture conforms to Raoult's law for the component in large excess and to Henry's law for the minor component. Find the Henry's law constants.



Method: Both Raoult's and Henry's laws are statements about the form of the graph of partial vapour pressure against mole fraction. Therefore, plot the partial vapour pressures against mole fraction.

Raoult's law is tested by comparing the data with the straight line $p_J = x_J p_J^*$ for each component in the region in which it is in excess (and acting as the solvent).

Henry's law is tested by finding a straight line $p_J = x_J K_J$ that is tangent to each partial vapour pressure at low x , where the component can be treated as the solute.



The experimental partial vapour pressures of a mixture of chloroform and acetone.

Answer: The data are plotted in the Figure together with the Raoult's law lines..

Henry's law requires
 $K = 175$ Torr for acetone and
 $K = 165$ Torr for chloroform



Using Henry's law

Estimate the molar solubility (the solubility in moles per liter) of oxygen in water at 25 °C and a partial pressure of 160 Torr, its partial pressure in the atmosphere at sea level.

$$K = 3.33 \times 10^7 \text{ Torr for oxygen in water at 298K}$$



Method: The mole fraction of solute is given by Henry's law as $x = p/K$, where p is the partial pressure of the gaseous solute. All we need do is to calculate the mole fraction that corresponds to the stated partial pressure, and then interpret that mole fraction as a molar concentration. For the latter part of the calculation, we calculate the amount of O_2 dissolved in 1.00 kg of water (which corresponds to about 1.00 L water). The solution is dilute, so the expressions for the mole fraction can be simplified.



Answer: Because the amount of O_2 dissolved is small, its mole fraction is

$$x(O_2) = \frac{n(O_2)}{n(O_2) + n(H_2O)} \approx \frac{n(O_2)}{n(H_2O)}$$

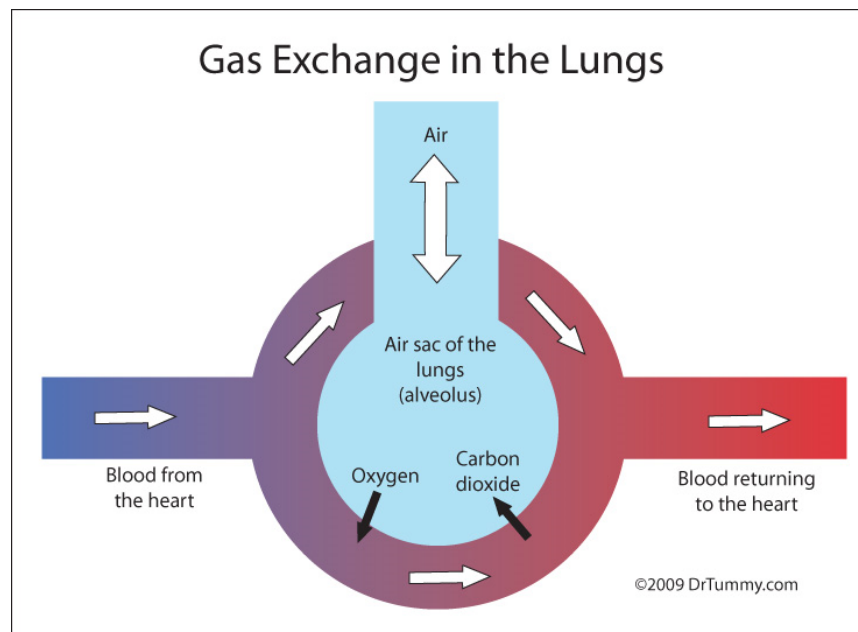
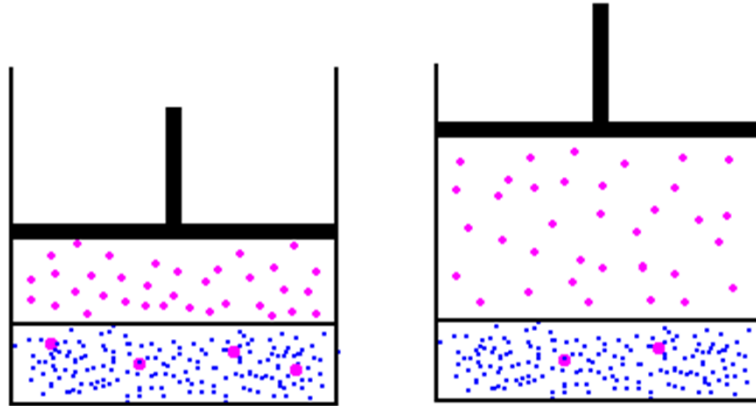
since $p_B = x_B K_B$ 

$$\begin{aligned} n(O_2) &\approx x(O_2)n(H_2O) = \frac{p(O_2)n(H_2O)}{K} \\ &\approx \frac{(160 \text{ Torr}) \times (55.5 \text{ mol})}{3.33 \times 10^7 \text{ Torr}} = 2.69 \times 10^{-4} \text{ mol} \end{aligned}$$

The molality of the saturated solution is therefore $2.69 \times 10^{-4} \text{ mol kg}^{-1}$, corresponding to a molar concentration of approximately $2.7 \times 10^{-4} \text{ mol L}^{-1}$



Henry's law applications





6.2 The properties of solutions

1. Colligative properties

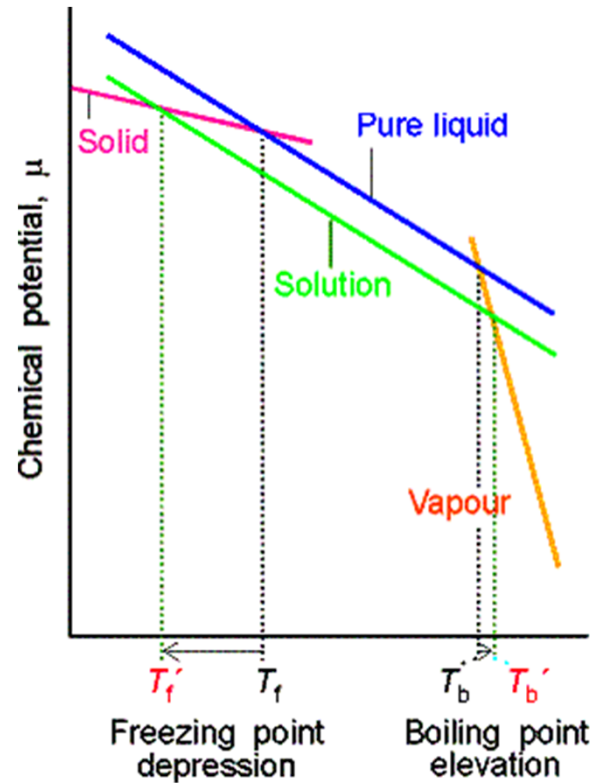
In dilute solutions, some properties depend only on the number of **solute** particles present, not their identity. They are called colligative properties.

Some of these properties are as followings:

- ❑ **The depression of vapour pressure**
- ❑ The elevation of boiling point
- ❑ The depression of freezing point
- ❑ Osmotic properties



a. The common features of colligative properties



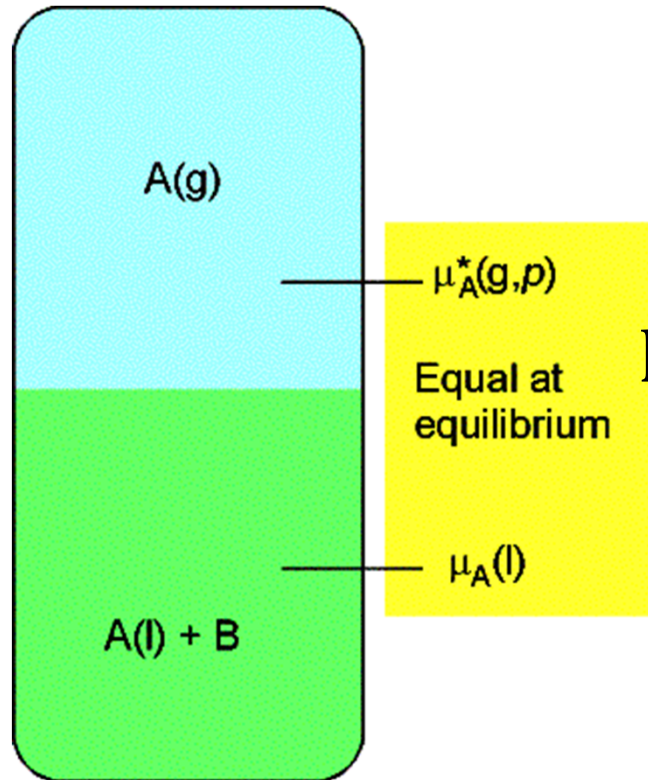
- Assume
 - Solute not volatile, non-electrolyte
 - Pure solute separates when frozen
- When you add solute the chemical potential, μ_A becomes

$$\mu_A = \mu_A^* + RT \ln x_A$$

Since $\ln x_A < 0$ $\mu_A < \mu_A^*$



b. The elevation of boiling point



$$\mu_A(l) = \mu_A^*(g)$$

$$\mu_A(l) = \mu_A^*(l) + RT \ln x_A$$

$$\ln(1 - x_B) = \frac{\mu_A^*(g) - \mu_A^*(l)}{RT} = \frac{\Delta_{vap} G_{m,A}}{RT}$$

$\Delta_{vap} G$: the Gibbs energy of vaporization of the pure solvent A.

x_B : the mole fraction of the solute.

$$\Delta_{vap} G = \Delta_{vap} H - T \Delta_{vap} S$$

Ignore the small temperature dependence of $\Delta_{vap} H$ and $\Delta_{vap} S$.

$$\ln(1 - x_B) = \frac{\Delta_{vap} H_{m,A}}{RT} - \frac{\Delta_{vap} S_{m,A}}{R}$$



When $x_B = 0$, the boiling point is that of pure liquid A, T^*

$$\ln 1 = 0 = \frac{\Delta_{vap}H_{m,A}}{RT} - \frac{\Delta_{vap}S_{m,A}}{R} \implies \frac{\Delta_{vap}H_{m,A}}{T^*} = \Delta_{vap}S_{m,A}$$

$$\ln(1 - x_B) = \frac{\Delta_{vap}H_{m,A}}{RT} - \frac{\Delta_{vap}S_{m,A}}{R} = \frac{\Delta_{vap}H_{m,A}}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

We now suppose that the amount of solute present is so small that $x_B \ll 1$, then $\ln(1 - x_B) \approx -x_B$

$$x_B = \frac{\Delta_{vap}H_{m,A}}{R} \left(\frac{1}{T^*} - \frac{1}{T} \right)$$

Finally, because $T \approx T^*$

$$\frac{1}{T^*} - \frac{1}{T} = \frac{T - T^*}{TT^*} \approx \frac{\Delta T}{T^{*2}}$$

$$\Delta T = K x_B \quad K = \frac{RT_A^{*2}}{\Delta_{vap}H_{m,A}}$$



The presence of a solute at a mole fraction x_B causes an increase in normal boiling point from T^* (boiling point of the pure solvent) to $T^* + \Delta T$ (boiling point of the dilute solution)

$$\Delta T = K x_B \quad K = \frac{RT_A^{*2}}{\Delta_{vap} H_{m,A}}$$

Because the equation make no reference to the identity of the solute, only to its mole fraction, we conclude that the elevation of boiling point is a colligative property. The value of ΔT does depend on the properties of the solvent, and the biggest changes occur for solvents with high boiling points.



For practical applications, we note that the mole fraction of B is proportional to its molality, b , in dilute solutions, then

$$\Delta T = K_b b$$

In dilute solution

$$x_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}} \approx \frac{n_{solute}}{n_{solvent}} = \frac{n_{solute} M_{solvent}}{m_{solvent}}$$

$$b = \frac{x_{solute}}{M_{solvent}} \quad K = \frac{RT_A^{*2} M_A}{\Delta_{vap} H_{m,A}}$$

K_b is the empirical ebullioscopic constant of the solvent



c. The depression of freezing point

The only difference between this calculation and the above is the appearance of the solid's chemical potential in place of the vapour's. we can write the result directly

$$\Delta T = K' x_B$$

$$K = \frac{RT_A^{*2}}{\Delta_{fus}H_{m,A}}$$

where ΔT is the freezing point depression, $\Delta_{fus}H$ is the enthalpy of fusion of the solvent; T^* is the freezing point of the pure solvent.

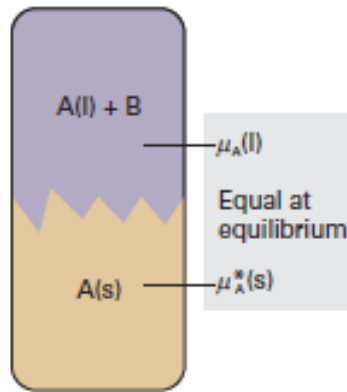


Fig. 5.24 The heterogeneous equilibrium involved in the calculation of the lowering of freezing point is between A in the pure solid and A in the mixture, A being the solvent and B a solute that is insoluble in solid A.

$$\mu_A(l) = \mu_A^*(s)$$

$$\mu_A(l) = \mu_A^*(l) + RT \ln x_A$$

$$\ln(1 - x_B) = \frac{\mu_A^*(s) - \mu_A^*(l)}{RT} = -\frac{\Delta_{fus}G_{m,A}}{RT}$$

$$\Delta T = T^* - T$$

$$\Delta T = K' x_B$$

$$K = \frac{RT_A^{*2}}{\Delta_{fus}H_{m,A}}$$



For practical applications, we note that the mole fraction of B is proportional to its molality, b , in dilute solutions, then

$$\Delta T = K_f b$$

K_f is the empirical cryoscopic constant of the solvent

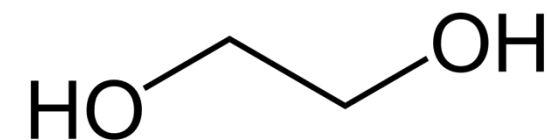
Table 7.2 Cryoscopic and ebullioscopic constants

	$K_f/(\text{K kg mol}^{-1})$	$K_b/(\text{K kg mol}^{-1})$
Acetic acid	3.90	3.07
Benzene	5.12	2.53
Camphor	40	
Carbon disulfide	3.8	2.37
Carbon tetrachloride	30	4.95
Naphthalene	6.94	5.8
Phenol	7.27	3.04
Water	1.86	0.51

Data: KL



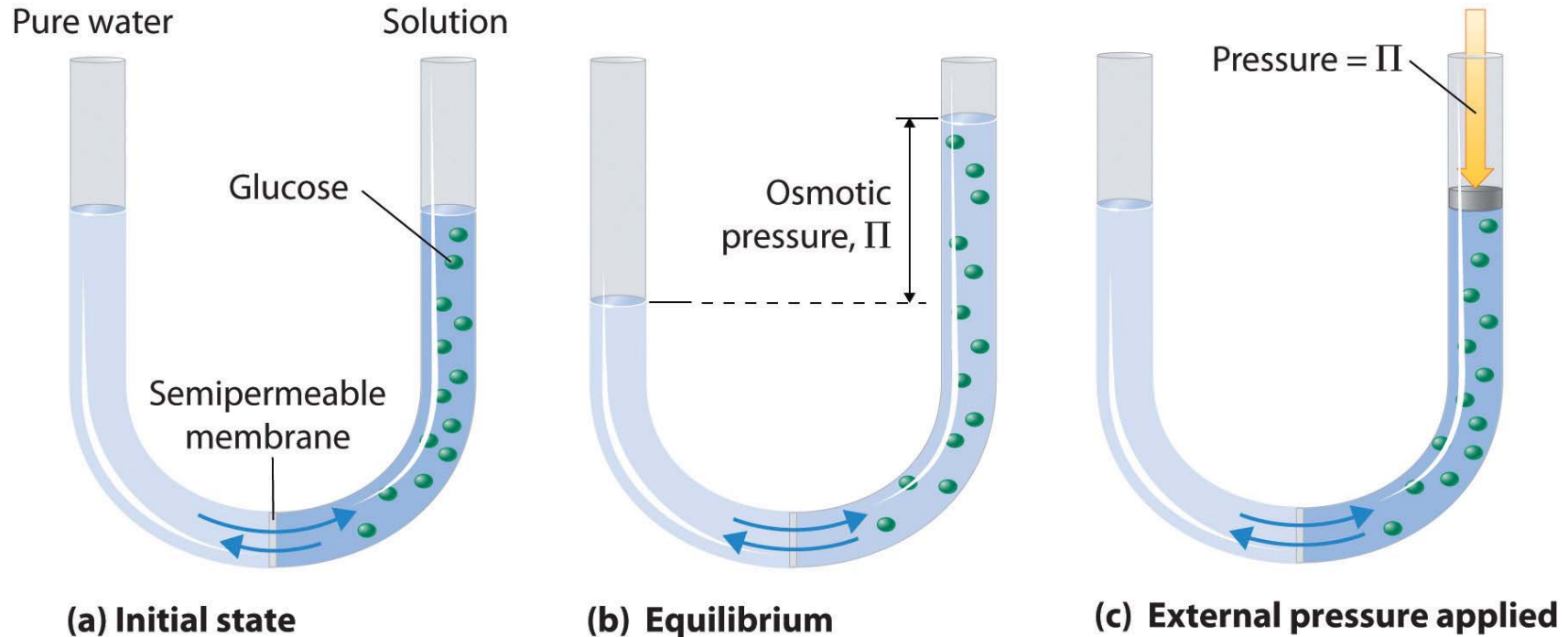
The salt sprinkled on this sidewalk makes the water on the sidewalk have a lower freezing point than pure water, so it does not freeze as easily. This makes walking on the sidewalk less hazardous in winter.



Ethylene glycol or antifreeze helps prevent the water in your vehicle's radiator from freezing through freezing-point depression. But you may not have realized that it also elevates the boiling point of the fluid as well. By raising the boiling point, it helps to protect against boil-overs. Many brands of antifreeze will list both the amount of boil-over protection and the freeze-up protection offered.



e. Osmosis



The phenomenon of osmosis is the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane, a membrane permeable to the solvent but not to the solute. The osmotic pressure, Π , is the pressure that must be applied to the solution to stop the influx of solvent.



J. A. Nollet (1748) - “wine spirits” in tube with animal bladder immersed in pure water

Semi-permeable membrane- water passes through into the tube.

Tube swells, sometimes bladder bursts

Increased pressure called osmotic pressure from Greek word meaning impulse

W. Pfeffer (1887)-quantitative study of osmotic pressure

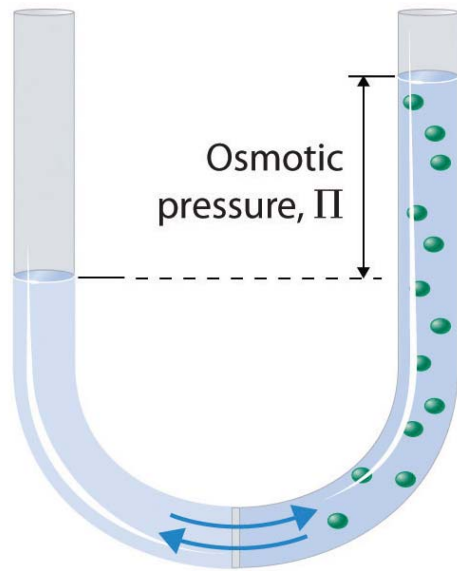
Membranes consisted of colloidal cupric ferrocyanide (氰亚铁酸铜)

Later work performed by applying external pressure to balance the osmotic pressure



On the pure solvent side the chemical potential of the solvent at pressure p , is $\mu_A^*(p)$

On the solution side the chemical potential is $\mu_A(x_A, p + \Pi)$



In equilibrium

$$\mu_A^*(p) = \mu_A(x_A, p + \Pi)$$

$$\mu_A(x_A, p + \Pi) = \mu_A^*(p + \Pi) + RT \ln x_A$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + \int_p^{p+\Pi} V_m dp$$

Where V_m is the molar volume of the pure solvent A

$$-RT \ln x_A = \int_p^{p+\Pi} V_m dp$$



For dilute solutions, $\ln x_A$ may be replaced by $\ln(1 - x_B) \approx -x_B$
We may also assume that the pressure range in the integration is so small that the V_m is a constant.

$$-RT \ln x_A = \int_p^{p+\Pi} V_m dp \quad RT x_B = \Pi V_m$$

$$\Pi = \frac{x_B}{V_m} RT \Rightarrow x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} \Rightarrow \Pi = \frac{n_B}{V} RT$$

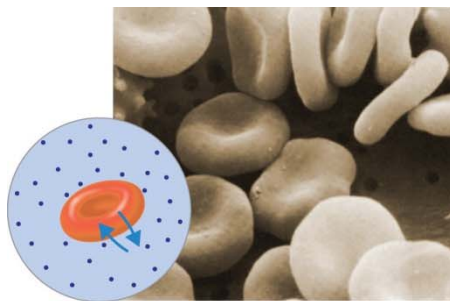
$$n_A V_m = V$$

$$\Pi = \frac{n_B}{V} RT$$

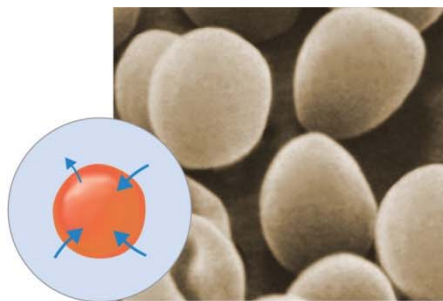
van't Hoff equation



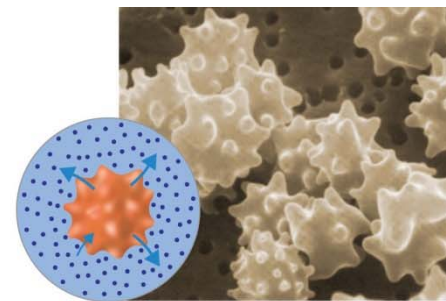
Effect on Red Blood Cells of the Surrounding Solution's Osmotic Pressure



(a) Cells in dilute salt solution



(b) Cells in distilled water



(c) Cells in concentrated salt solution

(a) When red blood cells are placed in a dilute salt solution having the same osmotic pressure as the intracellular fluid, the rate of flow of water into and out of the cells is the same and their shape does not change. (b) When cells are placed in distilled water whose osmotic pressure is less than that of the intracellular fluid, the rate of flow of water into the cells is greater than the rate of flow out of the cells. The cells swell and eventually burst. (c) When cells are placed in a concentrated salt solution with an osmotic pressure greater than that of the intracellular fluid, the rate of flow of water out of the cells is greater than the rate of flow into the cells. The cells shrivel and become so deformed that they cannot function.



Diffusion - The random movement of molecules from a area of higher concentration to an area of lower concentration (the net movement of molecules down a concentration gradient)

Osmosis - The diffusion of a solvent (water) across a selectively permeable membrane. In this case the water molecules always move from the weaker concentration (watery-hypotonic) to the stronger (hypertonic).

Dialysis - The diffusion of a solute (small molecules) across a selectively permeable membrane. In this case the solute molecules always move from the stronger concentration (hypertonic) to the weaker (hypotonic).



A patient undergoing hemodialysis depends on osmosis to cleanse the blood of waste products that the kidneys are incapable of removing due to disease.

The main function of the kidneys is to filter the blood to remove **wastes and extra water**, which are then expelled from the body as urine. Some diseases rob the kidneys of their ability to perform this function, causing a buildup of waste materials in the bloodstream. If a kidney transplant is not available or desirable, a procedure called dialysis can be used to remove waste materials and excess water from the blood.

In one form of dialysis, called *hemodialysis*, a patient's blood is passed through a length of tubing that travels through an *artificial kidney machine* (also called a **dialysis machine**). A section of tubing composed of a semipermeable membrane is immersed in a solution of sterile water, glucose, amino acids, and certain electrolytes. The osmotic pressure of the blood forces waste molecules and excess water through the membrane into the sterile solution. Red and white blood cells are too large to pass through the membrane, so they remain in the blood. After being cleansed in this way, the blood is returned to the body.



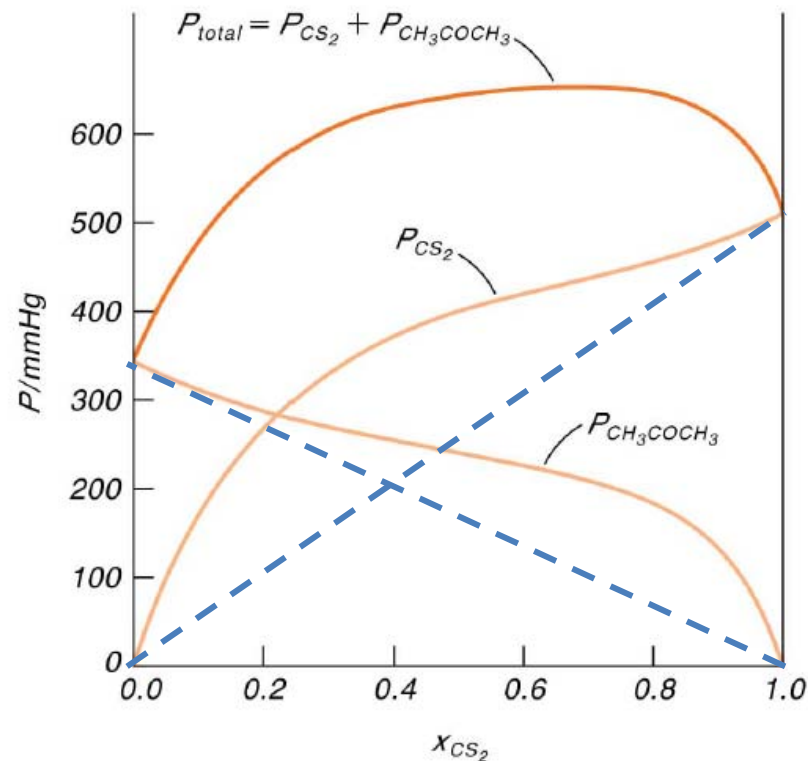
Colligative properties for nonvolatile solutions: Take it to the bank

- Vapour pressure is always lower
- Boiling point is always higher
- Freezing point is always lower
- Osmotic pressure



6.3 Activities

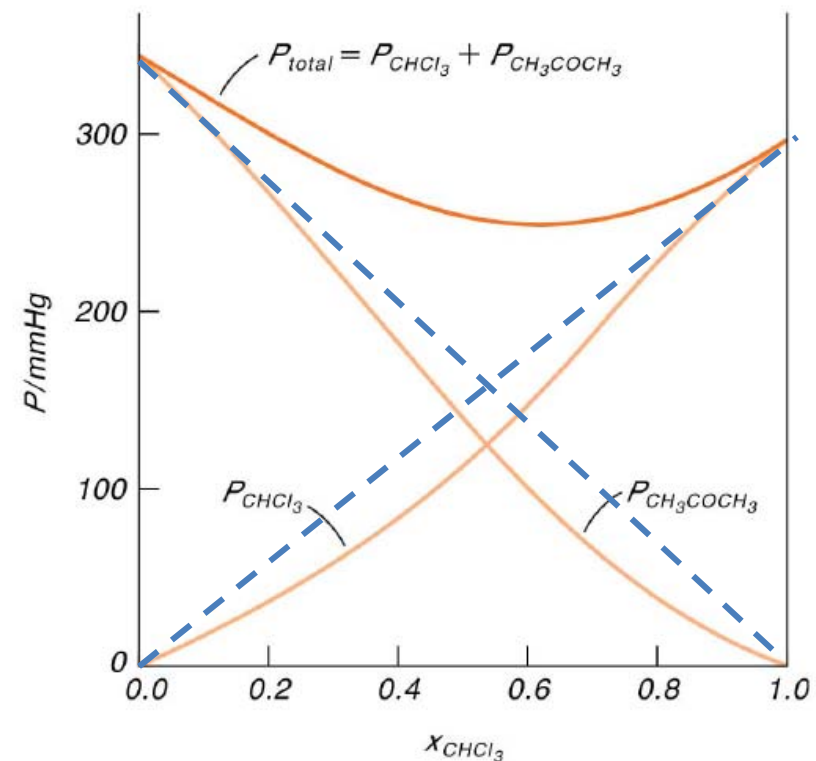
1. Real solutions



$$P_A > P_A^* x_A \quad (a)$$

$$P_B > P_B^* x_B$$

Positive deviation from Raoult's Law



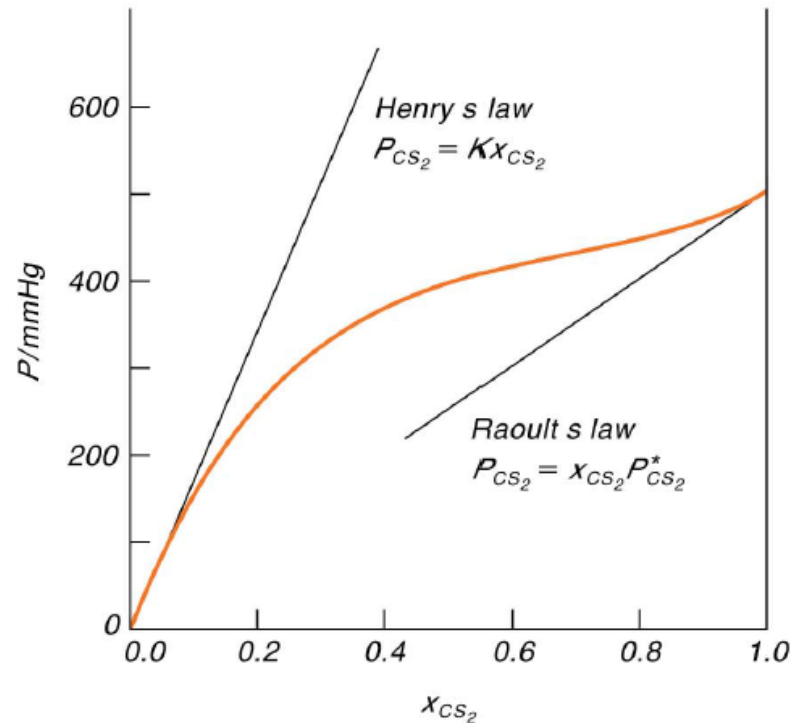
$$P_A < P_A^* x_A \quad (b)$$

$$P_B < P_B^* x_B$$

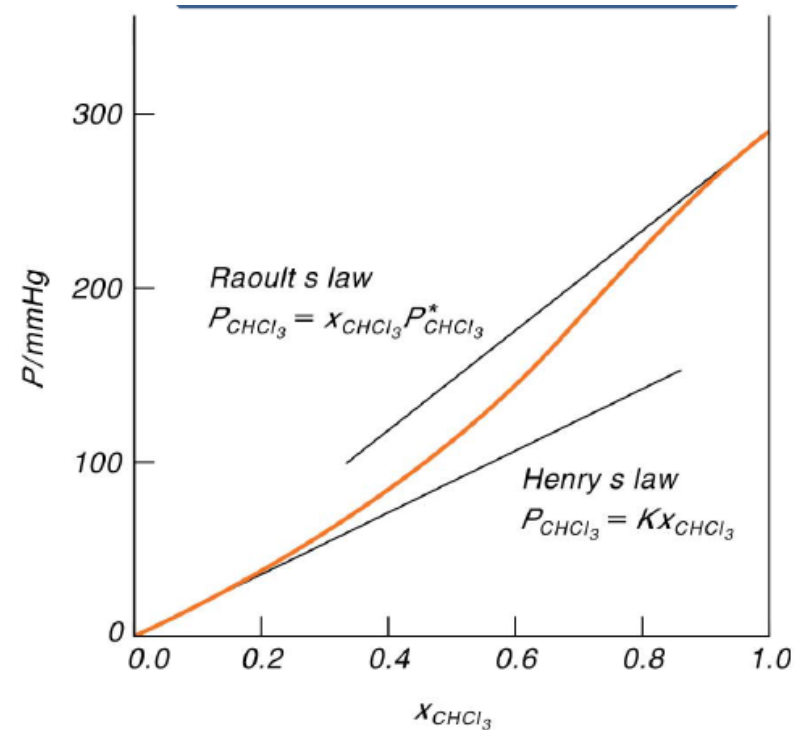
Negative deviation from Raoult's Law



Ideal-dilute solutions



(a)



(b)

Raoult's Law is a good approximation for solution components in excess:

Solvent $p_A = x_A p^*$

For low mole fraction components, Henry's Law is more appropriate:

Solute $p_B = x_B K_B$



2. The solvent activity---Derivation from Raoult's law

➤ Ideal solutions

$$\mu_A(\text{sln}) = \mu_A^*(l) + RT \ln x_A$$

$$x_A = \frac{p_A}{p_A^*} \Rightarrow \text{Raoult's law}$$

➤ Real solutions

$$x_A \neq \frac{p_A}{p_A^*} = \gamma_A x_A = a_A$$

$$\mu_A(\text{sln}) = \mu_A^*(l) + RT \ln \gamma_A x_A = \mu_A^*(l) + RT \ln a_A$$

a_A : the activity of A

γ_A : the activity coefficient



➤ The significances of a_A

$a_A = \gamma_A x_A$ Effective concentration (Corrected concentration)

$$\lim_{x_A \rightarrow 1} a_A = x_A$$

➤ The significances of γ_A

- ① $\gamma_A < 1$ positive derivation
- ② $\gamma_A > 1$ negative derivation
- ③ $\gamma_A = 1$ no derivation

$|\gamma_A - 1|$ The degree of derivation from Raoult's law
the degree of non-ideality of the solution

$$\lim_{x_A \rightarrow 1} \gamma_A = 1$$



➤ Standard chemical potential for a real solvent

$$\mu_A(\text{sln}) = \mu_A^*(l) + RT \ln a_A$$

$$\mu_A^\ominus: a_A = 1; T, p^\ominus$$

$$\mu_A^*(l): \text{Pure A liquid } \{T, p, x_A = 1\}$$

$$\mu_A^*(l) = \mu_A^\ominus + \int_{p^\ominus}^p V_{m,A} dp$$

$$\mu_A(\text{sln}) = \mu_A^\ominus + RT \ln a_A + \int_{p^\ominus}^p V_{m,A} dp$$

Generally, the effect of pressure on the properties of condensed matters can be negligible.

$$\mu_A(\text{sln}) = \mu_A^\ominus + RT \ln a_A$$



3. The solute activity---Derivation from Henry's law

➤ The chemical potential of solute B in a solution

$$\mu_B(\text{sln}) = \mu_B(g) = \mu_B^\ominus(g) + RT \ln \frac{p_B}{p^\ominus}$$

Solute B satisfies Henry's law: $p_B = K_B x_B$

$$\mu_B(\text{sln}) = \mu_B^\ominus(g) + RT \ln \frac{K_B}{p^\ominus} + RT \ln x_B$$



Solute B as a real solute

$$x_B \neq \frac{p_B}{K_B} = \gamma_B x_B = a_B$$

$$\begin{aligned}\mu_B(sln) &= \mu_B^\ominus(g) + RT \ln \frac{K_B}{p^\ominus} + RT \ln \gamma_B x_B \\ &= \mu_B^\ominus(g) + RT \ln \frac{K_B}{p^\ominus} + RT \ln a_B\end{aligned}$$



- Standard chemical potential for a real solute

$$\mu_B(\text{sln}) = \mu_B^\ominus(g) + RT \ln \frac{K_B}{p^\ominus} + RT \ln a_B$$

Standard state

if $x_B = 1$, $\gamma_B = 1$; so $a_B = 1$; T, K_B

A hypothetical state obeys Henry's law even at $x_B = 1$

$$\mu_B(\Delta) = \mu_B^\ominus(g) + RT \ln \frac{K_B}{p^\ominus}$$

$$\mu_B(\text{sln}) = \mu_B(\Delta) + RT \ln a_B$$



$$\mu_B(\text{sln}) = \mu_B^\ominus(\Delta) + RT \ln a_B$$

- The significances of a_B

$a_B = \gamma_B x_B$ Effective concentration (Corrected concentration)

$$\lim_{x_B \rightarrow 0} a_B = x_B$$

- The significances of γ_A

$|\gamma_B - 1|$ The degree of derivation from Henry's law
the degree of non-ideality of the solute

$$\lim_{x_B \rightarrow 0} \gamma_B = 1$$



➤ Solute activity in terms of molalities

For ideal-dilute solution $n_B \ll n_A$

$$p_B = K_{x,B} x_B \approx K_{x,B} \frac{n_B}{n_A} = K_{x,B} \frac{n_B M_A}{m_A} = K_{m,B} b_B$$

$$K_{m,B} = K_{x,B} M_A$$

$$\mu_B(\text{sln}) = \mu_B^\ominus(g) + RT \ln \frac{K_{m,B}}{p^\ominus} + RT \ln b_B$$



For real solute

$$K_{m,B} = K_{x,B} M_A$$

$$\mu_B(sln) = \mu_B^\ominus(g) + RT \ln \frac{K_{m,B}}{p^\ominus} + RT \ln a_B$$

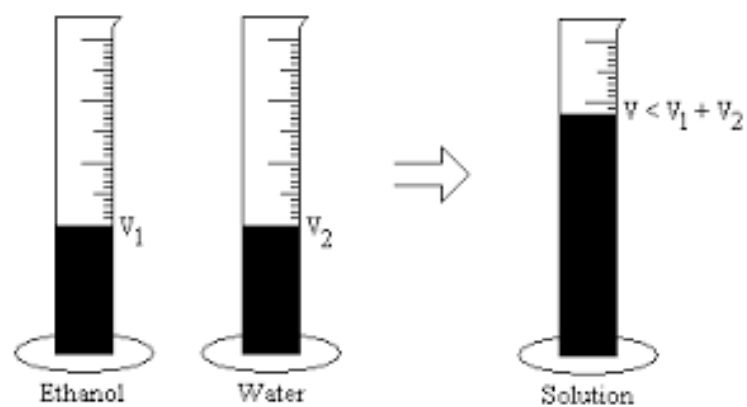
$$a_B = \gamma_B \frac{b_B}{b^\ominus}$$

Standard state $\gamma_B = 1 \quad b_B = b^\ominus = 1 \text{ mol kg}^{-1} \quad a_B = 1$

$$\mu_B(T, K_{m,B}, 1 \text{ mol kg}^{-1}) = \mu_B^\ominus(g) + RT \ln \frac{K_{m,B}}{p^\ominus}$$

p obeys Henry's law, $p_B = K_{m,B} b_B$

$$\mu_B(sln) = \mu_B(\blacksquare) + RT \ln a_B = \mu_B(\blacksquare) + RT \ln \gamma_B \frac{b_B}{b^\ominus}$$





Summary

1. Partial molar quantities

➤ definition

$$X_B = \left(\frac{\partial X}{\partial n_B} \right)_{T, p, n_A} \quad dX = X_A dn_A + X_B dn_B$$

➤ Additive formula

$$X = X_A n_A + X_B n_B \quad \text{One state at certain } T, p, x_A$$



To calculate an extensive state function **V; U; H; S; A; G**

Pure compound a: $X_{m,a}$ at n_a mol $\Rightarrow X = n_a \times X_{m,a}$

Pure compound b: $X_{m,b}$ at n_b mol $\Rightarrow X = n_b \times X_{m,b}$

If compound a and b make a mixture $X \neq n_a \times X_{m,a} + n_b \times X_{m,b}$

The contribution of 1 mol a/b to X is not $X_{m,a}/X_{m,b}$

In the mixture, the contribution of 1 mol a/b to X is dependent on x_a ($x_b = 1 - x_a$), in another word, composition.

Therefore, a new quantity has been induced. $\Rightarrow$

Partial molar quantity $X_a = \left(\frac{\partial X}{\partial n_a} \right)_{T, p, n_b}$ $X_b = \left(\frac{\partial X}{\partial n_b} \right)_{T, p, n_a}$ X_a or $X_b = f(T, p, x_a)$

$$X = n_a \times X_a + n_b \times X_b$$

$$X = f(T, p, n_a, n_b)$$



➤ Gibbs-Duhem equation

$$n_A d\mu_A + n_B d\mu_B = 0$$

2. Partial molar Gibbs energies= chemical potential

$$dG = Vdp - SdT + \mu_A dn_A + \mu_B dn_B \quad \mu_B = \left(\frac{\partial G}{\partial n_B} \right)_{T, p, n_A}$$

$$dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B \quad \mu_B = \left(\frac{\partial U}{\partial n_B} \right)_{S, V, n_A}$$

$$dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B \quad \mu_B = \left(\frac{\partial H}{\partial n_B} \right)_{S, p, n_A}$$

$$dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B \quad \mu_B = \left(\frac{\partial A}{\partial n_B} \right)_{T, V, n_A}$$



➤ Equilibrium criterion of chemical potential

$$(dG)_{T,p} = \sum_{i=1}^k \mu_i dn_i \quad \begin{array}{l} < 0 \text{ Spontaneous} \\ = 0 \text{ Equilibrium} \end{array}$$

The driving force for mass transfer is typically a difference in chemical potential.



3.The Gibbs energy of mixing

➤ General gases mixing at constant T

$$\Delta G_{mix} = n_A RT \ln \frac{p_A}{p^*(A)} + n_B RT \ln \frac{p_B}{p^*(B)}$$

➤ Perfect Gases/ideal solutions mixing at constant T, p and V ($V_A + V_B = V$)

$$\Delta G_{mix} = nRT(x_A \ln x_A + x_B \ln x_B) < 0$$

$$\Delta_{mix}S = -nR(x_A \ln x_A + x_B \ln x_B) > 0$$

$$\Delta_{mix}H = 0$$



4. Raoult's law and Henry's law

Raoult's

$$p_A = x_A p_A^*$$

Majority
Solvent

Minor modification

Henry's

$$p_B = x_B K_B \quad K_B \neq p_B^*$$

Minority
Solute

Major modification

Dilute
solution

Ideal solutions: Mixtures that obey the **Raoult's law** through out the composition range from pure A to pure B.

Ideal-dilute solutions: Mixtures for which the **solute obeys Henry's law** and the **solvent obeys Raoult's law**.



5. Colligative properties

- Vapour pressure is always lower

$$p_A = x_A p_A^*$$

- Boiling point is always higher

$$\Delta T = K_b b$$

$$\Delta T = T - T^*$$

K_b : the empirical ebullioscopic constant of the solution

- Freezing point is always lower

$$\Delta T = K_f b$$

$$\Delta T = T^* - T$$

K_f : the empirical cryoscopic constant of the solution

- Osmotic pressure drives solvent from lower concentration to higher concentration

$$\Pi = \frac{n_B}{V} RT$$



6. Chemical potential of gases

➤ Perfect gases

- pure $\mu_B = \mu_B^\ominus(T) + RT \ln \frac{p}{p^\ominus}$
- mixture $\mu_B = \mu_B^\ominus(T) + RT \ln \frac{p_B}{p^\ominus}$

➤ real gases

- pure $\mu_B = \mu_B^\ominus(T) + RT \ln \frac{f}{p^\ominus}$
- mixture $\mu_B = \mu_B^\ominus(T) + RT \ln \frac{f_B}{p^\ominus}$

$$\underline{\mu_B^\ominus(T) \quad B = f(\text{pure perfect gas}, T, p^\ominus = 1 \text{ bar})}$$



The fugacity coefficient, ϕ , is defined as:

$$\phi = \frac{f}{p} \quad \text{or} \quad f = \phi p$$

The fugacity coefficient, ϕ :

- 1). Dimensionless;
- 2). depends on the identity of the gas, the pressure, and the temperature.



7. Chemical potential of solvents

➤ Ideal solutions

$$\mu_B(l) = \mu_B^*(l) + RT \ln x_B = \mu_B^\ominus + RT \ln x_B$$

$$\underline{\mu_B^\ominus} = f(\text{pure liquid}, T, p^\ominus)$$



➤ Real solutions

Different meanings for different conditions

General Expression: $\mu_A = \mu_A^\ominus + RT \ln a_A$

- **Solvent** $\mu_A = \mu_A^\ominus + RT \ln a_A = \mu_A^\ominus + RT \ln \gamma_A x_A$

$$\mu_A^\ominus \quad A = f(\text{pure liquid}, T, p^\ominus)$$

Obey Raoult's law

- **Solute** $\mu_B = \mu_B(\Delta) + RT \ln a_B = \mu_B + RT \ln \gamma_B x_B$

$$\mu_B \quad B = f(\text{pure}, T, K_B)$$

$$\mu_B = \mu_B(\blacksquare) + RT \ln a_B = \mu_B + RT \ln \gamma_B \frac{b_B}{b^\ominus}$$

$$\mu_B \quad B = f(b_B = 1 \text{ mol kg}^{-1}, T, K_{m,B})$$

Obey Henry's law

Hypothetical state